INTERNSHIP AND PLACEMENT TRACKING SYSTEM

A PROJECT REPORT

Submitted by

TEJASRI M

(2116240701557)

in partial fulfilment of the End Semester Examination for the course

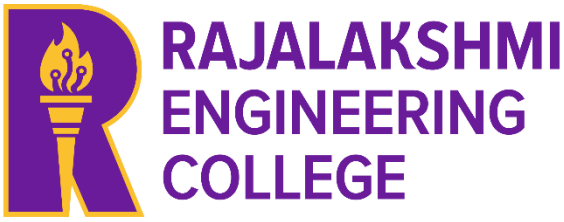
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "INTERNSHIP AND PLACEMENT TRACKING SYSTEM" is the bonafide work of TEJASRI M (2116240701557), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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DECLARATION

I, TEJASRI M (2116240701557), hereby declare that the project report titled "INTERNSHIP AND PLACEMENT TRACKING SYSTEM", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

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TEJASRI M

ABSTRACT

Training and placement cells in engineering colleges handle hundreds of students, dozens of recruiting companies and a steady stream of internship and placement opportunities every year. In most institutions this work is still managed through e-mail, notice boards, messaging groups, Google Forms and spreadsheets, so eligibility checks are manual, application status is hard to track, interview schedules and offers are scattered across files, and placement statistics are compiled by hand at the end of the semester.

The Internship and Placement Tracking System is a web-based application developed on the Zoho Creator low-code platform that provides a centralised platform for student information, company details, internship and placement opportunities, student applications, interview schedules, internship progress and placement offers through eight interconnected forms. Deluge scripts and workflows check eligibility, initialise application and interview statuses, track internship completion from weekly logs and manage placement offer status. An approval process lets the placement coordinator approve or reject each application, and a Blueprint drives the application through the stages Applied, Under Review, Shortlisted, Interview Scheduled, Selected and Rejected. Scheduled tasks send daily interview reminders and weekly and monthly summaries, and a dashboard shows registered students, companies, open opportunities, applications, interviews and offers. Access is controlled through four roles: Administrator (Placement Officer), Placement Coordinator, Student and Internship Mentor.

The application was verified through form-level, workflow-level and role-based test cases and is accessible on the web and on the Zoho Creator mobile app. It reduces manual administrative work, enforces eligibility rules consistently, gives students and administrators real-time visibility of every application, and supports better coordination between students and the placement cell.

Keywords: *Low-Code Development; Zoho Creator; Deluge Scripting; Blueprint; Approval Workflow; Placement Management*

PROJECT AT A GLANCE

Attribute	Details
Project Title	Internship and Placement Tracking System
Application Domain	Education – Training, Internship and Placement Management
Student Name & Register No.	Tejasri M – 2116240701557
Zoho Creator Edition	Zoho Creator (free trial edition, India data centre)
Application Name (Link Name)	Internship and Placement Tracking System (my-first-creator-app)
Published / Live URL	https://creatorapp.zoho.in/thendral.v.2024.cse_rajal553/my-first-creator-app/
Demo Credentials (per role)	Administrator: application owner account (tejasri.m.2024.cse@rajalakshmi.edu.in) | Placement Coordinator, Student and Internship Mentor roles are defined in Users and Permissions
Number of Forms / Subforms	8 / 2
Number of Reports (by type)	List: 8 Kanban: 1 Calendar: 1 Pivot/Chart: 3 Others: 0
Number of Pages / Dashboards	1 (Tracker Dashboard)
Number of Workflows	Form: 6 Schedule: 3 Approval: 1 Blueprint: 1 Batch: 0
Deluge Functions / Custom Actions	1 function (notifyStudent) / Blueprint transition and approval actions
Integrations	Zoho Mail (sendmail notifications); CSV/XLSX data import
Roles & Profiles Defined	Administrator (Placement Officer), Placement Coordinator, Student, Internship Mentor
Environments Used	Single application workspace; separate Stage and Production environments were not configured
Date of Production Deployment	Not separately deployed; live application URL used for demonstration

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CGPA	Cumulative Grade Point Average
CRUD	Create, Read, Update, Delete
CSV	Comma-Separated Values
CTC	Cost to Company
ER	Entity–Relationship
FR	Functional Requirement
IPTS	Internship and Placement Tracking System
KPI	Key Performance Indicator
LCAP	Low-Code Application Platform
LPA	Lakhs Per Annum
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
REST	Representational State Transfer
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

The Training and Placement Cell of an engineering college connects students with industry. Each year it registers several hundred final- and pre-final-year students from departments such as CSE, IT, ECE, EEE and Mechanical, invites recruiting companies, publishes internship and placement opportunities, checks which students are eligible for each opportunity, collects applications, arranges aptitude, technical and HR interviews, records offers and follows up on students' internships until a completion certificate is received.

The people involved are the Placement Officer, who administers the process; faculty Placement Coordinators, who verify applications and coordinate drives for their departments; Internship Mentors, who monitor students' internship work; and the students themselves. At present each group uses its own tools — e-mail, notice boards, messaging groups, Google Forms and spreadsheets — so information is duplicated and the status of a student or a drive is difficult to know at any moment. The Internship and Placement Tracking System (IPTS) described in this report brings this entire process into a single Zoho Creator application.

1.2 Problem Statement

Internship and placement activities in the institution are managed through disconnected e-mails, forms and spreadsheets, with no central record of students, companies, opportunities, applications, interviews, internship progress and offers. As a result, eligibility is checked manually, students and administrators cannot see the current status of an application, and placement statistics require slow manual compilation. The specific pain points are:

- **Manual eligibility screening:** CGPA, backlog and department criteria are checked by hand for every applicant, which is slow and error-prone.
- **No visibility of application status:** students repeatedly ask coordinators where their application stands, and there is no single record of each stage.

- **Scattered interview and offer data:** interview schedules, results and offers live in separate spreadsheets and messages, leading to missed interviews and inconsistent records.
- **Untracked internships:** internship progress and completion certificates are followed up through e-mails and paper logbooks.
- **Manual reporting and no access control:** weekly and semester placement statistics are compiled by hand, and spreadsheets containing student data are shared without role-based restrictions.

1.3 Objectives

- To design a normalised data model and forms for students, companies, internship and placement opportunities, applications, interviews, internship progress and offers.
- To automate eligibility checks, application and interview status initialisation, internship completion tracking, offer status management and periodic notifications using Deluge scripting, workflows and schedules.
- To manage the application lifecycle through an approval process and a Blueprint, and to provide role-based dashboards and reports for students, companies, opportunities, applications, interviews and offers.
- To integrate the application with Zoho Mail for automated notifications to students and administrators, and to import existing student lists through Zoho Creator's import facility.
- To deploy the application securely to a production environment with governance controls.

1.4 Scope of the Project

In scope: four roles — Administrator (Placement Officer), Placement Coordinator, Student and Internship Mentor — and eight modules: Student Profile, Company Profile, Internship Opportunity, Placement Opportunity, Student Application, Interview Schedule, Internship Progress and Placement Offer. The application covers eligibility validation, an approval process and a Blueprint for applications, interview scheduling and results, weekly internship logs with completion tracking, offer management, daily,

weekly and monthly scheduled notifications, list, Kanban, calendar, chart and pivot reports, a dashboard, and access on the web and the Zoho Creator mobile app.

Out of scope: a portal for recruiters to log in directly, online aptitude tests, résumé parsing, integration with the college ERP or external job portals, and predictive (AI-based) placement analytics. These are listed as future enhancements in Chapter 9.

1.5 Justification for a Low-Code Approach

A traditional (pro-code) implementation of IPTS would require a database schema for eight related entities, a backend with APIs, a responsive web front end, a mobile app, a workflow engine for approvals and stage transitions, a scheduler for reminders, and authentication with role-based access — several weeks of work for a team with diverse skills. Low-code application platforms (LCAPs) provide most of these capabilities as configurable building blocks [1].

Zoho Creator fits this problem closely. Forms generate the data model, with relationships through lookups and subforms; reports and the Page builder give list, Kanban, calendar, chart, pivot and dashboard views; Deluge expresses the eligibility rules and status logic; and built-in approval processes, Blueprints and schedules implement the recruitment lifecycle without custom workflow code [2]. Hosting, user management, audit trails and an auto-generated mobile app are included. Because the placement rules change every season (new eligibility criteria, new stages), the ability for staff to adjust forms and workflows without a development team is a key advantage of enterprise low-code platforms [3].

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing system the placement cell works as follows:

- A company contacts the placement cell by e-mail or phone with an internship or placement requirement.
- The placement officer circulates the opportunity through e-mail, notice boards and departmental messaging groups.
- Interested students register through a Google Form or send their résumés by e-mail.
- Coordinators download the responses and manually check each student's CGPA, backlogs and department against the eligibility criteria in a spreadsheet.
- Shortlists and interview schedules are shared as spreadsheets or messages; results and offers are recorded in separate workbooks.
- Internship progress is followed up through mentor e-mails and paper logbooks, and placement statistics are compiled manually at the end of the semester.

Figure 2.1 shows this as-is process flow.

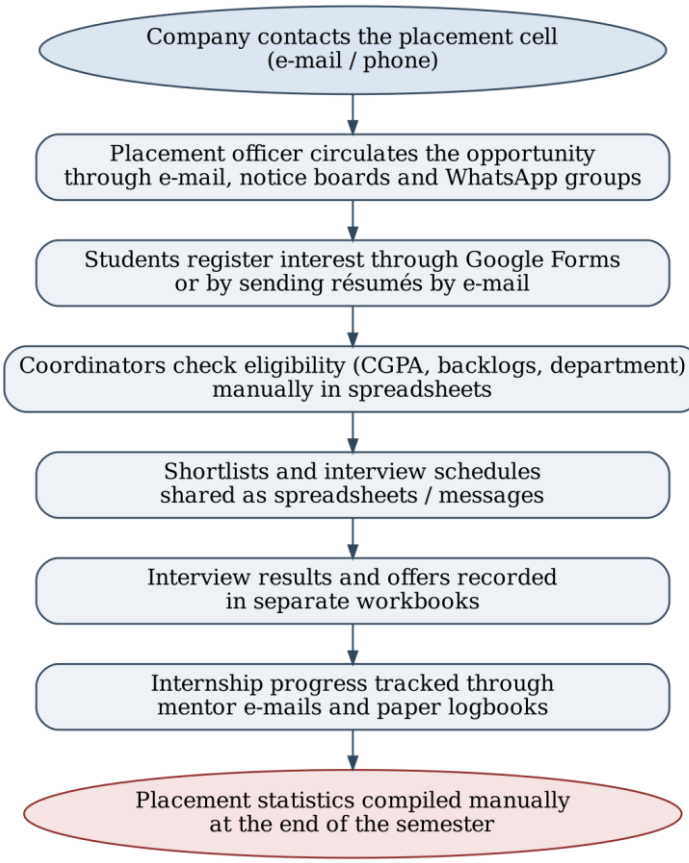


Figure 2.1: As-Is Process Flow of the Existing Placement Process

2.2 Limitations of the Existing System

- Student, company and opportunity data is duplicated across forms, spreadsheets and e-mails, with no single source of truth.
- Eligibility screening is manual, so ineligible or duplicate applications slip through and eligible students are sometimes missed.
- There is no defined, enforced sequence of application stages, and students cannot see the current status of their applications.
- Interview reminders, deadline closures and status updates depend on individual effort; internship completion is not tracked systematically.
- Weekly and semester statistics are compiled by hand, and shared spreadsheets expose student data without access control or a change history.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Approximately 10–12 weeks for eight entities, workflow engine, scheduler, web and mobile UI	Approximately 3–4 weeks: forms, reports, approval and Blueprint configured visually; Deluge for rules
Skills required	SQL, backend framework, front-end framework, mobile development, workflow/scheduler libraries, DevOps	Understanding of the placement process, Zoho Creator builders and basic Deluge scripting
Hosting & infrastructure	Server or cloud set-up, database administration, backups and SSL	Hosted and maintained by Zoho; backups, scaling and SSL handled by the platform
Mobile application	Separate mobile app or responsive UI must be built	Zoho Creator mobile app generated automatically
Security & access control	Authentication, RBAC, record-level rules and audit logging coded by hand	Built-in roles, permission profiles, record criteria and audit trail
Maintenance & change	Code change and redeployment for each new criterion, field or stage	Fields, criteria, approval steps and Blueprint stages changed through builders
Customisation limits	Practically unlimited UI and logic customisation	Limited UI customisation; logic bound by Deluge and platform limits
Cost	Developer effort plus hosting and maintenance cost	Per-user platform subscription; no infrastructure cost

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	A. Sahay, A. Indamutsa, D. Di Ruscio and A. Pierantonio, 2020 [4]	Supporting the understanding and comparison of low-code development platforms	Taxonomy of low-code platform features (data modelling, business logic, workflow, interoperability, deployment) with a comparison of eight platforms.	Used to identify the platform capabilities IPTS must exercise, especially workflow and data modelling.

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
2	R. Sanchis, Ó. García-Perales, F. Fraile and R. Poler, 2020 [5]	Low-code as enabler of digital transformation in manufacturing industry	Low-code shortens delivery of business applications and lets domain experts build and maintain them.	Supports building IPTS so that placement-cell staff can maintain it themselves.
3	M. Hirzel, 2023 [6]	Low-code programming models (Communications of the ACM)	Classifies low-code programming models and discusses where visual configuration stops and code is needed.	Guided the split between approval/Blueprint configuration and Deluge scripts (Section 8.5).
4	Gartner, Inc., 2024 [3]	Magic Quadrant for Enterprise Low-Code Application Platforms	Low-code platforms are becoming mainstream for enterprise and departmental applications.	Justifies a low-code approach for an institutional process application.
5	Zoho Corporation [7]	Zoho Recruit (applicant tracking system)	Commercial tool with job openings, candidate pipelines, interview scheduling and offer management.	Benchmark for pipeline stages and interview handling; IPTS adds eligibility rules, internship tracking and campus-specific roles.

2.5 Emerging Trends in Low-Code Development

Several low-code trends shaped IPTS [8]. **Process automation with visual workflows** — approval processes and Blueprints — allowed the recruitment lifecycle to be modelled as configuration rather than code. **Hyper-automation** encouraged automating every repetitive step: eligibility checks, status initialisation, reminders and periodic summaries. **Citizen development** led to a design in which placement staff can change criteria and stages themselves, with Deluge limited to short, well-commented scripts. **Mobile-first access** was met by the Zoho Creator mobile app, which students use most. **AI-assisted development** (Zia) was studied, and résumé keyword extraction and placement prediction are planned as future enhancements (Section 9.2).

2.6 Proposed System

The proposed system is a single Zoho Creator application with eight interconnected forms, an approval process, a Blueprint, form workflows, schedules and a dashboard. It overcomes each limitation of Section 2.2 as follows:

- All student, company and opportunity data is stored once and referenced through lookups.
- On-validate scripts reject duplicate, late and ineligible applications automatically.
- The approval process and Blueprint enforce a defined sequence of stages, and students see their status in My Applications.
- Scheduled workflows send interview reminders, close expired opportunities and track internship completion from weekly logs.
- Weekly and monthly summaries, charts and the dashboard replace manual statistics, while roles, record-level criteria and the audit trail protect student data.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator (Placement Officer)	Head of the placement cell; owns the application	Manage users and roles, student and company profiles, publish opportunities, record offers, view all data, dashboard and summaries
Placement Coordinator	Faculty member coordinating placements for a department	Approve or reject applications, move applications through Blueprint stages, schedule interviews and record results
Student	Pre-final or final-year student	Maintain own profile, apply for opportunities, track application status, view interviews, accept or decline offers, submit internship logs
Internship Mentor	Faculty member guiding students during internships	Review weekly internship logs, add remarks and verify completion certificates

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall maintain Student Profiles with unique register number and e-mail, department, CGPA, active backlogs, skills and résumé, and allow bulk import.	Administrator	High
FR-02	The system shall maintain Company Profiles with industry, type, website, location and HR contacts (subform).	Administrator	High
FR-03	The system shall allow internship opportunities to be published with eligible departments, minimum CGPA, stipend, duration and application deadline.	Administrator, Coordinator	High
FR-04	The system shall allow placement opportunities to be published with CTC, eligible departments, minimum CGPA, maximum backlogs, deadline and drive date.	Administrator, Coordinator	High
FR-05	The system shall reject duplicate applications, applications after the deadline and applications from ineligible students.	Student, System	High

FR ID	Requirement	Role(s)	Priority
FR-06	The system shall initialise each new application to Applied / Pending approval and e-mail an acknowledgement to the student.	System	High
FR-07	The system shall route every application to a Placement Coordinator for approval or rejection with remarks, and notify the student.	Coordinator	High
FR-08	The system shall enforce the stage sequence Applied → Under Review → Shortlisted → Interview Scheduled → Selected / Rejected through a Blueprint.	Coordinator	High
FR-09	The system shall allow interviews only for shortlisted applications, initialise their status to Scheduled / Pending, notify the student and mark them Completed when a result is entered.	Coordinator, System	High
FR-10	The system shall compute internship completion from weekly logs and mark the internship Completed when all weeks are logged and a certificate is uploaded.	Student, Mentor, System	Medium
FR-11	The system shall allow offers only for Selected placement applications, manage the offer status (Offered / Accepted / Declined) and mark the student Placed on acceptance.	Administrator, Student	High
FR-12	The system shall send daily interview reminders, close expired opportunities, and e-mail weekly and monthly summaries.	System	Medium
FR-13	The system shall display a dashboard and reports of students, companies, opportunities, applications, interviews and offers.	Administrator, Coordinator	High
FR-14	The system shall restrict each role to permitted forms and reports; students shall see only their own records.	All	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports with up to 2,000 records shall load within 3 seconds; form submission with validation shall complete within 2 seconds.
NFR-02	Security	Students shall access only their own profile, applications, interviews, progress and offers; academic fields shall be read-only for students.
NFR-03	Usability	All student actions shall be possible on the Zoho Creator mobile app; forms shall show only relevant fields through layout rules.
NFR-04	Availability	The application shall be available 24 × 7 on Zoho's cloud, subject to Zoho's service availability.
NFR-05	Maintainability	Eligibility criteria, stages and approvers shall be changeable through the builders; e-mail logic shall be centralised in one function.
NFR-06	Auditability	All record changes, approvals and Blueprint transitions shall be traceable through the audit trail and history.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator, free trial edition (India data centre)
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing tool	Not required – no external REST API is used in this version
Other Zoho services	Zoho Mail (e-mail notifications through the sendmail task)
Mobile	Zoho Creator mobile app (Android / iOS)
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

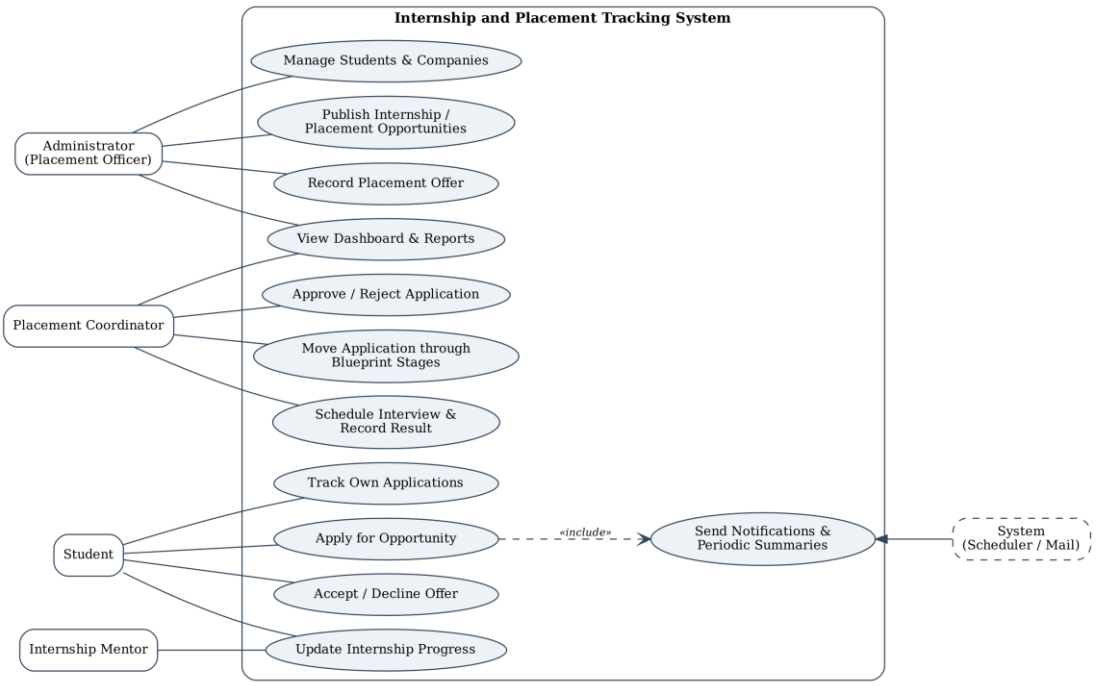


Figure 3.1: Use Case Diagram of the Internship and Placement Tracking System

Table 3.5: Use Case Description – Apply for a Placement Opportunity

Use Case ID	UC-01
Use Case Name	Apply for a Placement Opportunity
Actor(s)	Student (primary); Placement Coordinator; System
Pre-condition	The student is logged in and has a complete profile; the placement opportunity is Open.
Main Flow	1. The student opens Placement Opportunities and selects a drive. 2. The student fills in the Student Application form and submits it. 3. DS-01 checks duplicate, deadline, CGPA, backlog and department criteria and sets status Applied / Pending. 4. DS-03 e-mails an acknowledgement. 5. The approval request reaches the Placement Coordinator, who approves it; the student is notified and the application enters the Blueprint at Applied.
Alternate Flow	3a. The student is ineligible, has already applied or the deadline has passed: an alert is shown and submission is cancelled. 5a. The coordinator rejects the application with remarks: status becomes Rejected and the student is notified.
Post-condition	The application is stored with Approval_Status Approved (or Rejected) and is visible in My Applications.
Related FR	FR-05, FR-06, FR-07

Table 3.6: Use Case Description – Process Application to Offer

Use Case ID	UC-02
Use Case Name	Process Application to Offer
Actor(s)	Placement Coordinator, Administrator, Student; System
Pre-condition	The application is approved and at the Applied stage of the Blueprint.
Main Flow	1. The coordinator executes Start Review and then Shortlist; DS-05 updates the status and e-mails the student. 2. The coordinator creates an Interview Schedule record; DS-06 sets it to Scheduled / Pending and DS-07 notifies the student. 3. After the final round the coordinator enters the result; DS-08 marks the interview Completed. 4. The coordinator executes Select. 5. The administrator records a Placement Offer; DS-11 sets it to Offered and the student accepts it; DS-12 marks the student Placed.
Alternate Flow	3a. Result is Not Cleared: the coordinator executes Reject and the student is notified. 5a. The student declines the offer: the placement officer is alerted.
Post-condition	Application is Selected or Rejected; the offer and the student's placement status are up to date.
Related FR	FR-08, FR-09, FR-11

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

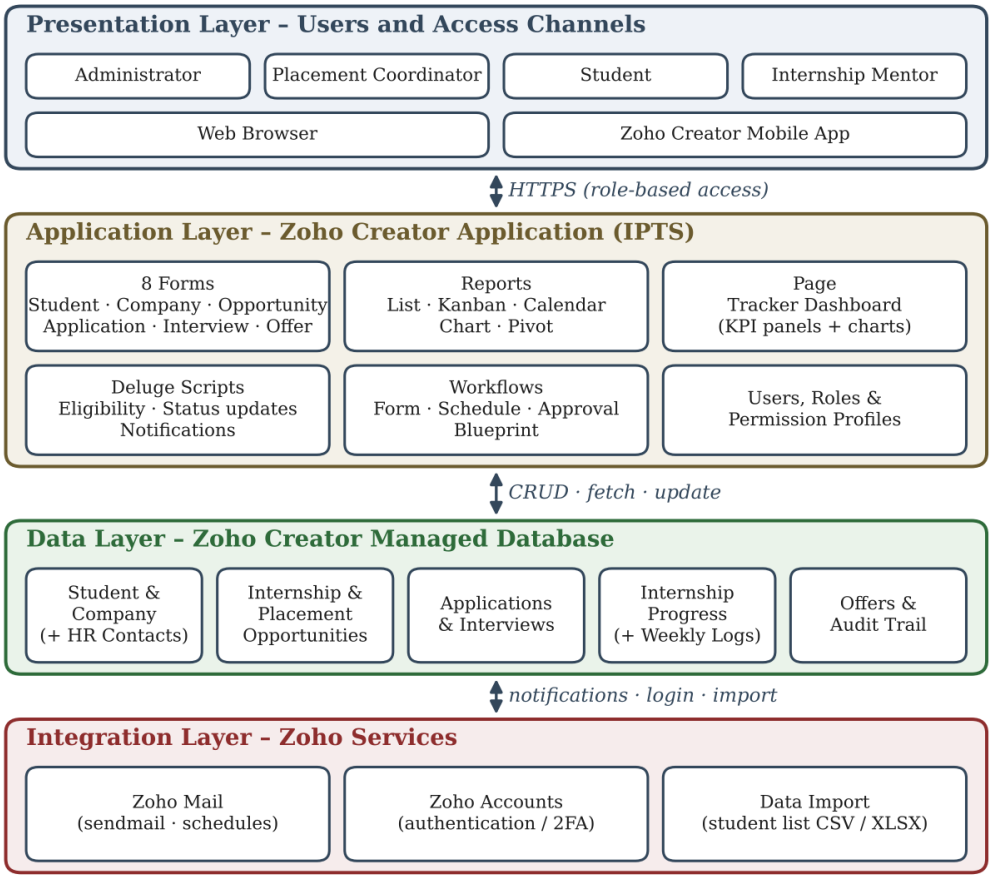


Figure 4.1: System Architecture of the Internship and Placement Tracking System

Figure 4.1 shows the four-layer architecture of IPTS. The **presentation layer** consists of the four roles, who access the application through a web browser or the Zoho Creator mobile app over HTTPS; each user sees only the sections permitted by his or her role. The **application layer** is the Zoho Creator application: eight forms capture data, reports and the dashboard present it, Deluge scripts implement eligibility and status logic, workflows (form workflows, schedules, the approval process and the Blueprint) drive the process, and roles and permission profiles enforce access.

The **data layer** is Zoho Creator's managed database, in which each form is a table and relationships are held through lookup fields and two subforms; the audit trail records every change. The **integration layer** contains the Zoho services used: Zoho Mail

delivers notifications and scheduled summaries, Zoho Accounts provides authentication, and the import facility loads the student list from the college's spreadsheets.

4.2 Data Model

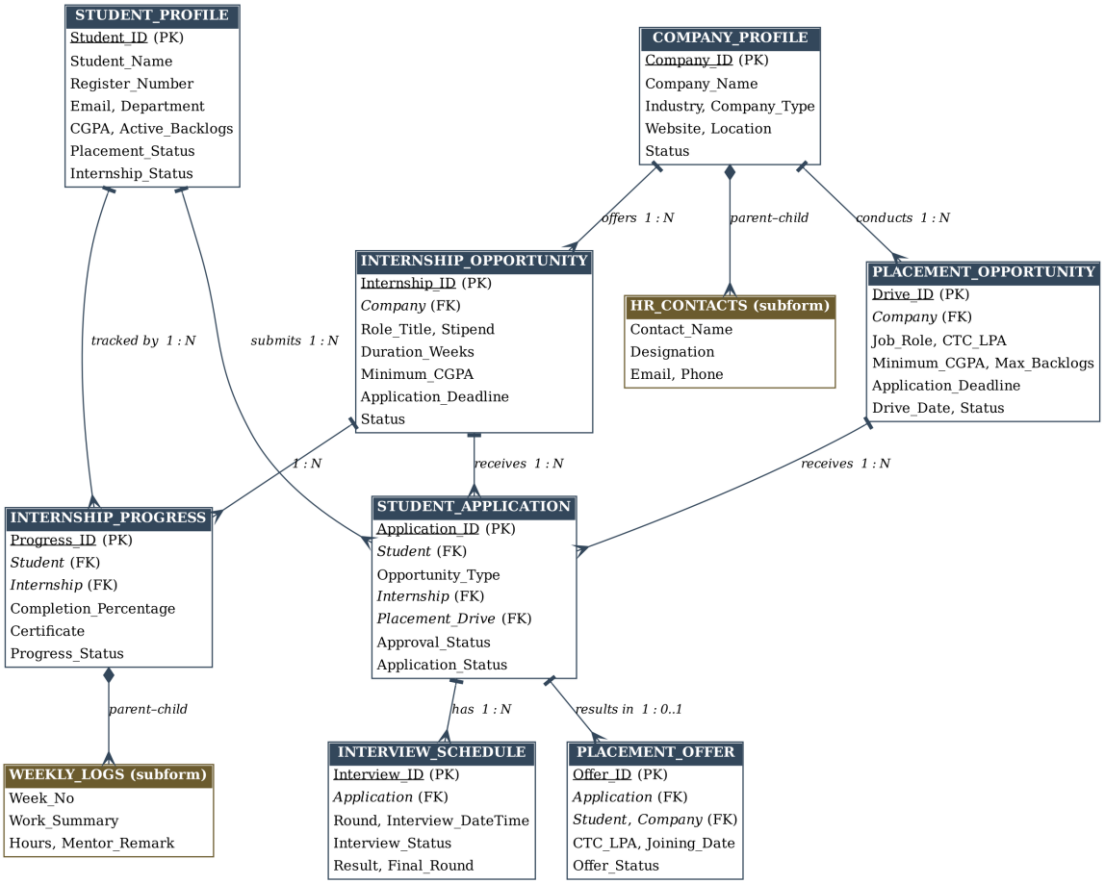


Figure 4.2: Entity–Relationship Diagram of IPTS

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Company Profile	Company_ID, Company_Name, Industry	HR Contacts	Parent–Child	Subform in Company Profile
Company Profile	Company_ID, Company_Name	Internship Opportunity	One-to-Many	Lookup (Company) in Internship Opportunity
Company Profile	Company_ID, Company_Name	Placement Opportunity	One-to-Many	Lookup (Company) in Placement Opportunity

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Student Profile	Student_ID, Register_Number, CGPA	Student Application	One-to-Many	Lookup (Student) in Student Application
Internship / Placement Opportunity	Internship_ID / Drive_ID	Student Application	One-to-Many	Lookups (Internship, Placement_Drive) in Student Application
Student Application	Application_ID, Application_Status	Interview Schedule	One-to-Many	Lookup (Application) in Interview Schedule
Student Application	Application_ID	Placement Offer	One-to-One (optional)	Lookup (Application) in Placement Offer
Student Profile	Student_ID	Internship Progress	One-to-Many	Lookup (Student) in Internship Progress
Internship Progress	Progress_ID, Completion_Percentage	Weekly Logs	Parent–Child	Subform in Internship Progress

Student Profile and Company Profile are the master entities. Each company offers internship and placement opportunities, and its HR contacts are stored as subform rows. Student Application is the central transactional entity: it links a student to exactly one internship or one placement opportunity (chosen by Opportunity_Type), and it in turn has interview records and, for selected placement applications, a placement offer. Internship Progress links a student to an internship and holds the weekly logs as subform rows, from which completion is computed (DS-09).

4.3 Form Design

Tables 4.2 to 4.9 specify the fields of the eight forms using Zoho Creator field types. Together they use more than eighteen distinct field types, including Lookup, Subform, Multi Select, Decision Box, Date-Time, Percent, URL, Address, Image and File Upload.

Table 4.2: Field Specification – Student Profile Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Student ID	Student_ID	Auto Number	—	Prefix STU-	System generated
Student Name	Student_Name	Name	Yes	First and last name	—
Register Number	Register_Number	Single Line	Yes	No duplicate values; 13 digits	—
Email	Email	Email	Yes	No duplicate values	Used for login criteria and notifications
Phone	Phone	Phone	Yes	10-digit number	—
Department	Department	Drop Down	Yes	—	CSE / IT / AIDS / ECE / EEE / MECH / …
Year of Passing	Year_of_Passing	Number	Yes	4 digits	—
CGPA	CGPA	Decimal	Yes	0.00–10.00	Read-only for students
Active Backlogs	Active_Backlogs	Number	Yes	Minimum 0; default 0	Read-only for students
Skills	Skills	Multi Select	No	—	Java, Python, Web, Data, …
Résumé	Resume	File Upload	Yes	PDF only	—
Placement Status	Placement_Status	Drop Down	—	Default: Not Placed	Updated by DS-12
Internship Status	Internship_Status	Drop Down	—	Default: Not Started	Updated by DS-10

Table 4.3: Field Specification – Company Profile Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Company ID	Company_ID	Auto Number	—	Prefix CMP-	System generated
Company Name	Company_Name	Single Line	Yes	No duplicate values	—
Industry	Industry	Drop Down	Yes	—	IT Services / Product / Core / Finance / …
Company Type	Company_Type	Radio	Yes	—	MNC / Start-up / PSU
Website	Website	URL	No	Valid URL	—

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Location	Location	Address	No	—	—
HR Contacts	HR_Contacts	Subform	No	—	Contact_Name, Designation, Email, Phone
Logo	Logo	Image	No	—	—
Status	Status	Radio	Yes	Default: Active	Active / Inactive

Table 4.4: Field Specification – Internship Opportunity Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Internship ID	Internship_ID	Auto Number	—	Prefix INT-	System generated
Company	Company	Lookup (Company Profile)	Yes	Only Active companies	—
Role / Title	Role_Title	Single Line	Yes	—	—
Description	Description	Rich Text	No	—	—
Mode	Mode	Radio	Yes	—	On-site / Remote / Hybrid
Duration (weeks)	Duration_Weeks	Number	Yes	1–52	Used by DS-09
Stipend	Stipend	Currency (INR)	No	Minimum 0	—
Eligible Departments	Eligible_Departments	Multi Select	Yes	—	Checked by DS-01
Minimum CGPA	Minimum_CGPA	Decimal	Yes	0.00–10.00	Checked by DS-01
Application Deadline	Application_Deadline	Date	Yes	Not in the past	—
Status	Status	Drop Down	Yes	Default: Open	Open / Closed (closed by DS-13)

Table 4.5: Field Specification – Placement Opportunity Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Drive ID	Drive_ID	Auto Number	—	Prefix PLC-	System generated

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Company	Company	Lookup (Company Profile)	Yes	Only Active companies	—
Job Role	Job_Role	Single Line	Yes	—	—
Job Description	Job_Description	Rich Text	No	—	—
CTC (LPA)	CTC_LPA	Decimal	Yes	Greater than 0	—
Eligible Departments	Eligible_Departments	Multi Select	Yes	—	Checked by DS-01
Minimum CGPA	Minimum_CGPA	Decimal	Yes	0.00–10.00	Checked by DS-01
Maximum Backlogs	Max_Backlogs	Number	Yes	Default 0	Checked by DS-01
Application Deadline	Application_Deadline	Date	Yes	≤ Drive Date	—
Drive Date	Drive_Date	Date	Yes	—	—
Status	Status	Drop Down	Yes	Default: Open	Open / Closed (closed by DS-13)

Table 4.6: Field Specification – Student Application Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Application ID	Application_ID	Auto Number	—	Prefix APP-	System generated
Student	Student	Lookup (Student Profile)	Yes	Defaults to logged-in student	—
Opportunity Type	Opportunity_Type	Radio	Yes	—	Internship / Placement
Internship	Internship	Lookup (Internship Opportunity)	Conditional	Status = Open	Shown when type = Internship
Placement Drive	Placement_Drive	Lookup (Placement Opportunity)	Conditional	Status = Open	Shown when type = Placement
Applied Date	Applied_Date	Date	—	Set by DS-01	Read-only
Cover Note	Cover_Note	Multi Line	No	—	—
Approval Status	Approval_Status	Drop Down	—	Default: Pending	Pending / Approved / Rejected

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Approval Remarks	Approval_Remarks	Multi Line	No	—	Entered by coordinator
Application Status	Application_Status	Drop Down	—	Set by DS-01 / Blueprint	Applied / Under Review / Shortlisted / Interview Scheduled / Selected / Rejected

Table 4.7: Field Specification – Interview Schedule Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Interview ID	Interview_ID	Auto Number	—	Prefix INV-	System generated
Application	Application	Lookup (Student Application)	Yes	Status = Shortlisted / Interview Scheduled	—
Round	Round	Drop Down	Yes	—	Aptitude / Technical / HR
Interview Date-Time	Interview_DateTime	Date-Time	Yes	In the future (DS-06)	—
Mode	Mode	Radio	Yes	—	Online / Offline
Venue / Meeting Link	Venue_or_Link	Single Line	Yes	—	—
Interview Status	Interview_Status	Drop Down	—	Set by DS-06 / DS-08	Scheduled / Completed / Cancelled
Result	Result	Drop Down	—	Default: Pending	Pending / Cleared / Not Cleared
Final Round	Final_Round	Decision Box	No	Default: unchecked	—
Feedback	Feedback	Multi Line	No	—	—

Table 4.8: Field Specification – Internship Progress Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Progress ID	Progress_ID	Auto Number	—	Prefix IPR-	System generated

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Student	Student	Lookup (Student Profile)	Yes	—	—
Internship	Internship	Lookup (Internship Opportunity)	Yes	—	—
Mentor Email	Mentor_Email	Email	Yes	—	Faculty mentor
Start Date	Start_Date	Date	Yes	—	—
End Date	End_Date	Date	Yes	≥ Start Date	—
Weekly Logs	Weekly_Logs	Subform	No	—	Week_No, Work_Summary, Hours, Mentor_Remark
Completion (%)	Completion_Percentage	Percent	—	Computed by DS-09	Read-only
Certificate	Certificate	File Upload	No	PDF / image	Required for completion
Progress Status	Progress_Status	Drop Down	—	Set by DS-09	Ongoing / Completed

Table 4.9: Field Specification – Placement Offer Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Offer ID	Offer_ID	Auto Number	—	Prefix OFR-	System generated
Application	Application	Lookup (Student Application)	Yes	Status = Selected (DS-11)	—
Student	Student	Lookup (Student Profile)	—	Auto-filled by DS-11	Read-only
Company	Company	Lookup (Company Profile)	—	Auto-filled by DS-11	Read-only
CTC (LPA)	CTC_LPA	Decimal	No	Defaults to drive CTC	—
Offer Date	Offer_Date	Date	Yes	Default: today	—
Joining Date	Joining_Date	Date	No	≥ Offer Date	—
Response Deadline	Response_Deadline	Date	Yes	≥ Offer Date	—
Offer Letter	Offer_Letter	File Upload	Yes	PDF only	—

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Offer Status	Offer_Status	Drop Down	—	Default: Offered (DS-11)	Offered / Accepted / Declined

4.4 Process and Workflow Design

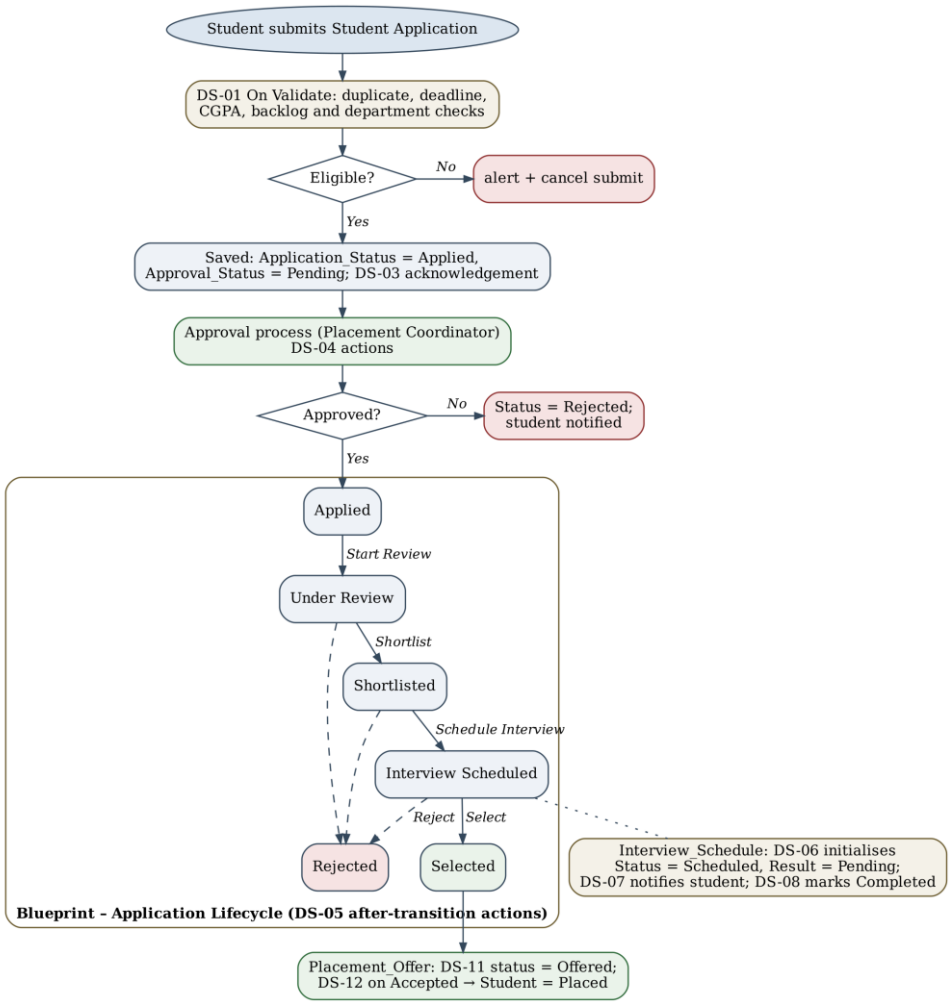


Figure 4.3: Application Workflow with Approval Process and Blueprint

Figure 4.3 shows the central process of IPTS. When a student submits an application, DS-01 checks for duplicates, the deadline and the CGPA, backlog and department criteria; ineligible applications are rejected with an alert. Valid applications are saved with status Applied and approval Pending, and DS-03 sends an acknowledgement. The approval process routes the application to a Placement Coordinator; a rejection closes it, while an approval lets it enter the Blueprint. The coordinator then moves it through Under Review, Shortlisted and Interview Scheduled to Selected or Rejected, and each

transition's after-action (DS-05) updates the status and informs the student. Interview records are initialised and closed by DS-06 to DS-08, and a selected placement application leads to a Placement Offer whose status is managed by DS-11 and DS-12. Separately, three schedules (DS-13 to DS-15) send daily reminders and weekly and monthly summaries.

4.5 User Interface and Navigation Design

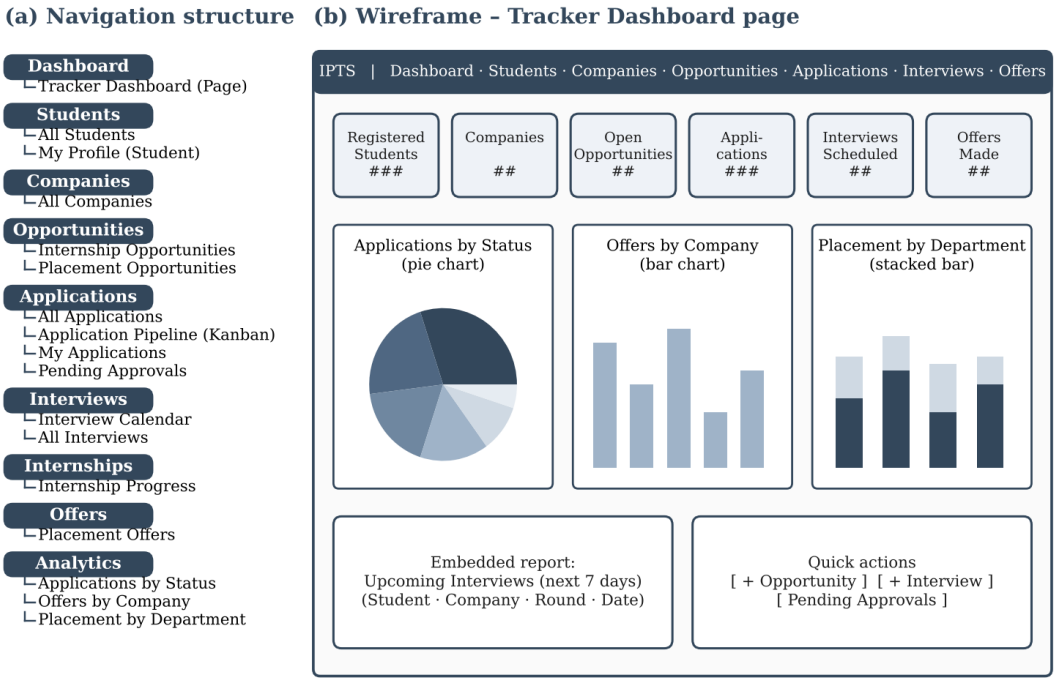


Figure 4.4: Navigation Structure and Wireframe of the Tracker Dashboard

The menu is organised into nine sections — Dashboard, Students, Companies, Opportunities, Applications, Interviews, Internships, Offers and Analytics — shown in Figure 4.4(a). Students see only the sections and reports that concern them (Opportunities, My Applications, Interviews, Internship Progress and Offers). The dashboard wireframe in Figure 4.4(b) places six KPI panels at the top, three charts in the middle and the list of upcoming interviews with quick-action buttons at the bottom.

4.6 Access Control Design

Table 4.10: Role–Permission Matrix

Component	Administrator	Placement Coordinator	Student	Internship Mentor
Form: Student Profile	A E V D	V	E V (own; academic fields read-only)	V (mentees)
Form: Company Profile	A E V D	V	V	—
Forms: Internship / Placement Opportunity	A E V D	A E V	V (Open only)	V
Form: Student Application	A E V D	E V (approve, transitions)	A V (own)	—
Form: Interview Schedule	A E V D	A E V	V (own)	—
Form: Internship Progress	A E V D	V	A E V (own)	E V (mentees)
Form: Placement Offer	A E V D	A E V	E V (own; status only)	—
Reports: Analytics charts & pivot	V	V	—	—
Page: Tracker Dashboard	V	V	—	—
Schedules, approval and Blueprint settings	Full	—	—	—

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application was created from scratch (blank application) in the Zoho Creator workspace and named “Internship and Placement Tracking System”. Forms were created in dependency order — Student Profile and Company Profile first, then the two opportunity forms, Student Application, Interview Schedule, Internship Progress and Placement Offer — so that every lookup could refer to an existing form. The existing student list was imported from the placement cell’s spreadsheet.

The menu was arranged into the nine sections designed in Section 4.5, with the Tracker Dashboard as the home page for staff and the Opportunities section as the home page for students. A professional theme and the college logo were applied for consistent branding.

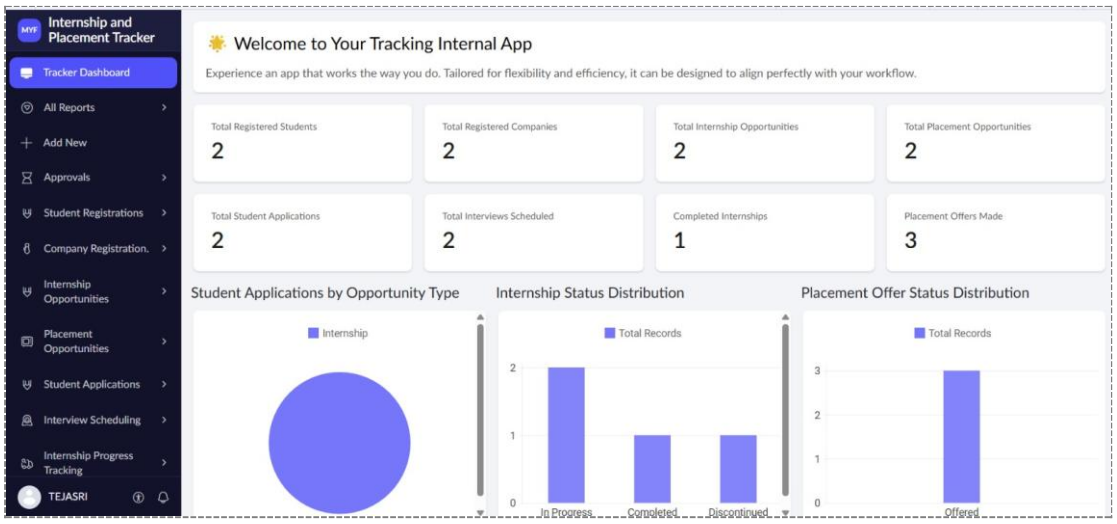


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

The eight forms were built with the fields specified in Tables 4.2–4.9. Key configuration choices were:

- **Lookups:** Company in both opportunity forms; Student, Internship and Placement_Drive in Student Application; Application in Interview Schedule and Placement Offer; Student and Internship in Internship Progress. Lookup filters show only Active companies and Open opportunities.

- **Subforms:** HR_Contacts in Company Profile and Weekly_Logs in Internship Progress.
- **Field validations:** unique register number, e-mail and company name; numeric ranges for CGPA, backlogs and duration; PDF-only file uploads; and read-only status fields maintained by scripts.
- **Layout rules:** in Student Application, the Internship lookup is shown only when Opportunity Type is Internship and the Placement Drive lookup only when it is Placement; in Interview Schedule, Feedback becomes mandatory once a Result is entered.
- **Form properties:** custom success messages (e.g., "Your application has been submitted for approval"), Submit and Reset buttons, and the Student field defaulted to the logged-in student.

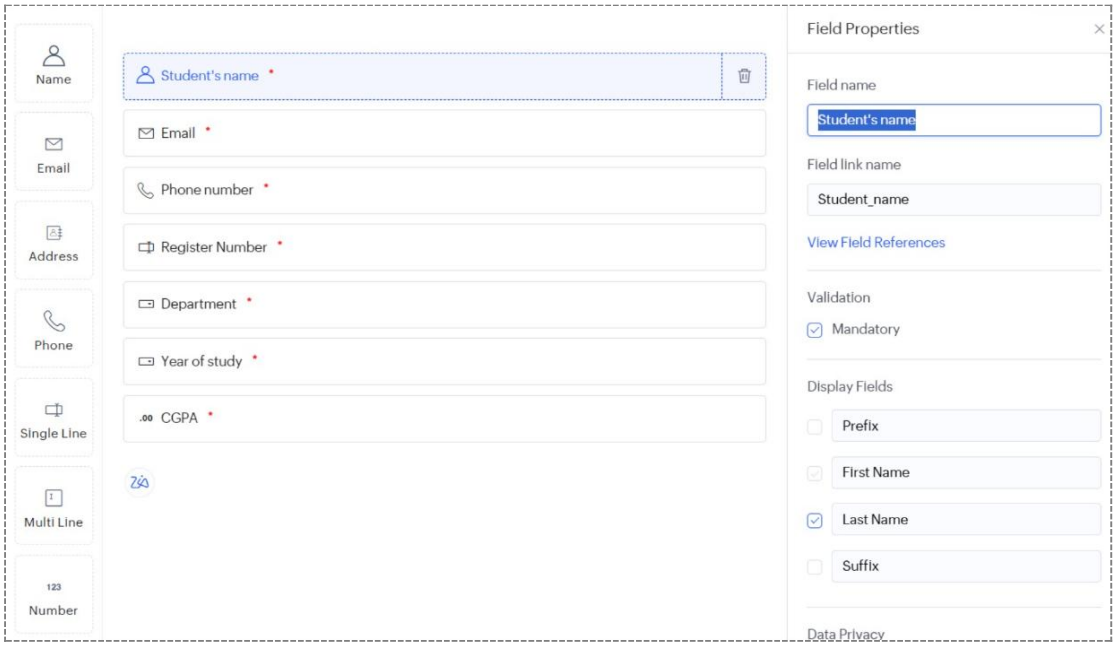


Figure 5.2: Student Application Form in the Form Builder

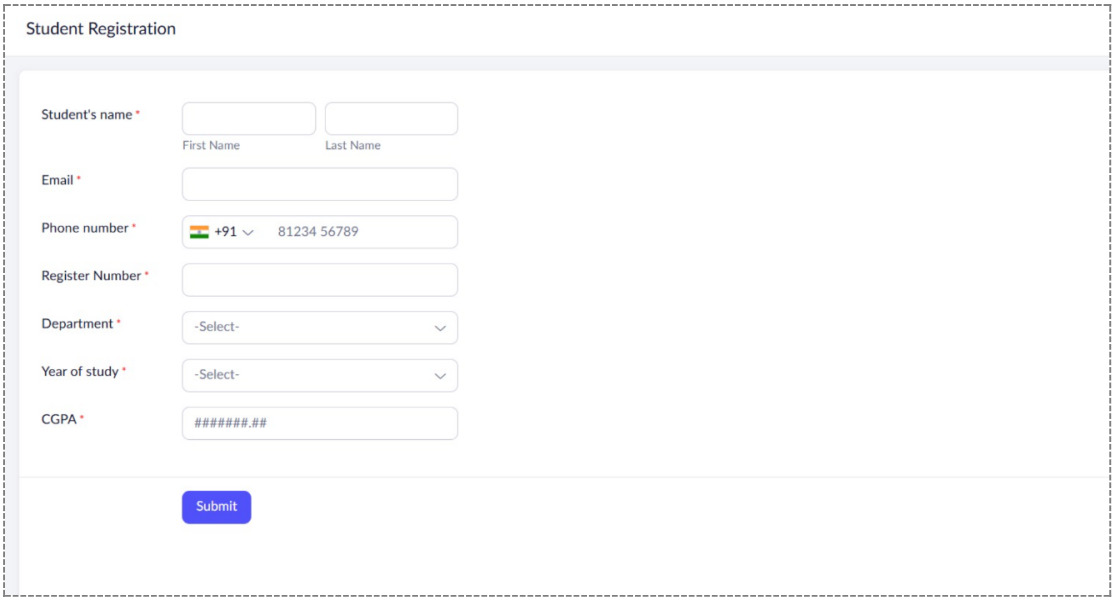


Figure 5.3: Student Application Form in Live Mode

5.3 Reports

Thirteen reports of five types were created (Table 5.1). Each list report has a quick view with the key columns and a detail view with related lists; for example, the detail view of an application shows its interviews and offer, and the detail view of a company shows its opportunities.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Students	List	Student Profile	Group by Department; filter by Placement Status; search by Skills	—	Admin, Coordinator
All Companies	List	Company Profile	Sort by Company Name; filter by Industry	—	Admin, Coordinator
Internship Opportunities	List	Internship Opportunity	Default filter Status = Open; sort by Application Deadline	—	All roles
Placement Opportunities	List	Placement Opportunity	Default filter Status = Open; sort by Drive Date	—	All roles

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Applications	List	Student Application	Group by Application Status; filter by Opportunity Type	Blueprint transition buttons; approve / reject	Admin, Coordinator
My Applications	List	Student Application	Criteria: Student.Email == zoho.loginuserid	—	Student
Application Pipeline	Kanban	Student Application	Columns by Application Status	—	Admin, Coordinator
Interview Calendar	Calendar	Interview Schedule	Interview Date-Time; colour by Round	—	Admin, Coordinator, Student (own)
Internship Progress	List	Internship Progress	Group by Progress Status; red when Completion < 50 % near End Date	—	Admin, Mentor, Student (own)
Placement Offers	List	Placement Offer	Group by Company; filter by Offer Status	—	Admin, Coordinator, Student (own)
Applications by Status	Chart (pie)	Student Application	Count grouped by Application Status	—	Admin, Coordinator
Offers by Company	Chart (bar)	Placement Offer	Count grouped by Company	—	Admin, Coordinator
Placement by Department	Pivot Table	Student Profile	Rows: Department; columns: Placement Status; value: count	—	Admin, Coordinator

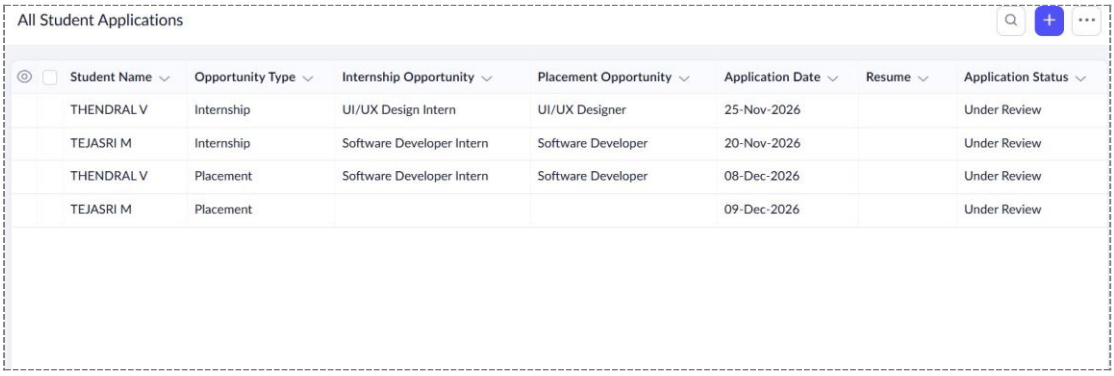


Figure 5.4: All Applications Report Grouped by Status

5.4 Pages and Dashboards

The **Tracker Dashboard** page was built with the Page builder for the Administrator and Placement Coordinator roles. It contains:

- **Six KPI panels:** Registered Students, Companies, Open Opportunities, Applications, Interviews Scheduled and Offers Made, each an aggregate over the corresponding form.
- **Three analytics components:** Applications by Status (pie chart), Offers by Company (bar chart) and Placement by Department (pivot).
- **An embedded report** of interviews in the next seven days, and **buttons** to add an opportunity, schedule an interview and open pending approvals.

Because every component reads live data, the dashboard replaces the manually compiled statistics of the existing system.

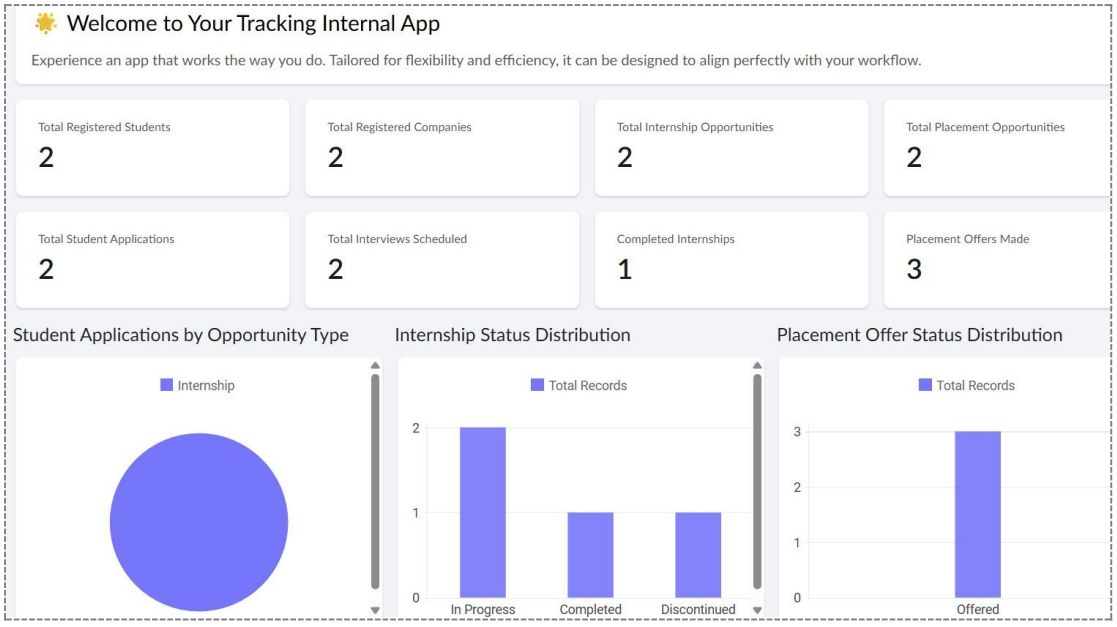


Figure 5.5: Tracker Dashboard Page

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Student Application → Validations on form submission (Created)	Reject duplicate, late and ineligible applications; initialise status	fetch, count(), contains(), if-else, alert, cancel submit
DS-02	Function: notifyStudent	Common e-mail to a student	function with arguments, sendmail

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-03	Student Application → Successful submission (Created)	Acknowledge the application	fetch, string concatenation, function call
DS-04	Approval process → On Approve / On Reject	Set approval and application status; notify	field update, ifnull, function call
DS-05	Blueprint → transition after-actions	Keep Application Status in step with the stage; notify	field update, function call
DS-06	Interview Schedule → Validations on form submission (Created)	Allow only shortlisted applications; initialise status	fetch, date-time comparison, alert
DS-07	Interview Schedule → Successful submission (Created)	Notify the student of the interview	fetch, function call
DS-08	Interview Schedule → Successful submission (Edited)	Mark Completed; alert coordinator after final round	record update, sendmail
DS-09	Internship Progress → Validations on form submission	Compute completion %; set Ongoing / Completed	subform loop, arithmetic, round()
DS-10	Internship Progress → Successful submission	Update the student's internship status	fetch, record update, function call
DS-11	Placement Offer → Validations on form submission (Created)	Allow only Selected placement applications; auto-fill; set Offered	fetch, lookup assignment
DS-12	Placement Offer → Successful submission (Created / Edited)	Manage offer status; mark student Placed	if-else-if, record update, sendmail
DS-13	Schedule: daily 08:00 AM	Interview reminders; close expired opportunities	criteria fetch, for each, addDay(), toDate()
DS-14	Schedule: weekly, Monday 09:00 AM	Weekly summary to the placement officer	count(), subDay(), sendmail
DS-15	Schedule: monthly, day 1, 09:00 AM	Monthly placement statistics	count(), sum(), round(), toString()

Code 5.1: Eligibility validation and status initialisation (DS-01)

```
// DS-01 | Form: Student_Application | Event: Validations on form
submission (Created)
// Purpose: Block duplicate, late and ineligible applications;
initialise status
stu = Student_Profile[ID == input.Student];
if(input.Opportunity_Type == "Internship")
{
    opp = Internship_Opportunity[ID == input.Internship];
    dup = Student_Application[Student == input.Student && Internship
== input.Internship].count();
}
else
{
    opp = Placement_Opportunity[ID == input.Placement_Drive];
    dup = Student_Application[Student == input.Student &&
Placement_Drive == input.Placement_Drive].count();
    if(stu.Active_Backlogs > opp.Max_Backlogs)
    {
        alert "Your active backlogs exceed the limit for this drive.";
        cancel submit;
    }
}
if(dup > 0)
{
    alert "You have already applied for this opportunity.";
    cancel submit;
}
if(opp.Status != "Open" || zoho.currentdate >
opp.Application_Deadline)
{
    alert "Applications for this opportunity are closed.";
    cancel submit;
}
if(stu.CGPA < opp.Minimum_CGPA)
{
    alert "A minimum CGPA of " + opp.Minimum_CGPA + " is required.";
    cancel submit;
}
if(!opp.Eligible_Departments.contains(stu.Department))
{
    alert "Your department is not eligible for this opportunity.";
    cancel submit;
}
input.Application_Status = "Applied";
input.Approval_Status = "Pending";
input.Applied_Date = zoho.currentdate;
```

Code 5.2: Internship completion tracking (DS-09)

```
// DS-09 | Form: Internship_Progress | Event: Validations on form
submission
// Purpose: Compute completion from weekly logs and set the progress
status
opp = Internship_Opportunity[ID == input.Internship];
weeksLogged = 0;
for each log in input.Weekly_Logs
{
    weeksLogged = weeksLogged + 1;
}
pct = (weeksLogged * 100.0 / opp.Duration_Weeks).round(0);
if(pct > 100)
{
    pct = 100;
}
input.Completion_Percentage = pct;
if(pct == 100 && input.Certificate != null && input.Certificate != "")
{
    input.Progress_Status = "Completed";
}
else
{
    input.Progress_Status = "Ongoing";
}
```

DS-01 runs before an application is saved. It fetches the student and the selected opportunity, counts existing applications by the same student for the same opportunity, and cancels the submission with a specific alert if the student has already applied, the opportunity is closed or past its deadline, the CGPA is below the minimum, the backlogs exceed the limit, or the department is not in the eligible list. Otherwise it initialises Application_Status to Applied, Approval_Status to Pending and the applied date. In testing, a student with CGPA 6.8 was blocked from a drive requiring 7.0, and a second application to the same drive was rejected (TC-06, TC-07).

DS-09 counts the rows of the Weekly_Logs subform, converts them to a percentage of the internship's duration (capped at 100 %), and sets Progress_Status to Completed only when all weeks are logged and a certificate is uploaded. DS-10 then copies the status to the student's profile and sends a congratulatory e-mail. For an eight-week internship with six logs the completion was 75 %; after the eighth log and the certificate upload the record became Completed (TC-15). The complete source of all fifteen scripts is given in Appendix C [9].

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Application Eligibility Check	Form Workflow	Validations on form submission (Created)	—	Block invalid applications; initialise status	DS-01
Application Acknowledgement	Form Workflow	Successful form submission (Created)	—	E-mail the student	DS-03, DS-02
Application Approval	Approval	Record creation	Approval_Status = Pending	Route to Placement Coordinator; approve / reject actions	DS-04
Application Lifecycle	Blueprint	Stage transitions	Approval_Status = Approved	Applied → Under Review → Shortlisted → Interview Scheduled → Selected / Rejected	DS-05
Interview Initialisation	Form Workflow	Validations and successful submission (Created)	—	Set Scheduled / Pending; notify student	DS-06, DS-07
Interview Result	Form Workflow	Successful form submission (Edited)	Result ≠ Pending	Mark Completed; alert after final round	DS-08
Internship Completion Tracking	Form Workflow	Validations and successful submission	—	Compute completion; update student	DS-09, DS-10
Offer Status Management	Form Workflow	Validations (Created) and successful submission	—	Set Offered; mark Placed on acceptance	DS-11, DS-12
Daily Reminders	Schedule	Every day, 08:00 AM	Interview tomorrow; deadline passed	Remind students; close opportunities	DS-13

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Weekly Summary	Schedule	Every Monday, 09:00 AM	—	Summary e-mail to placement officer	DS-14
Monthly Report	Schedule	Day 1 of every month, 09:00 AM	—	Statistics e-mail to placement officer	DS-15

Approval process. "Application Approval" is triggered when an application is created. It is assigned to users with the Placement Coordinator role, who see the request in the Pending Approvals list and approve or reject it with remarks. The On Approve and On Reject actions (DS-04) set the approval and application status and notify the student.

Blueprint. "Application Lifecycle" is defined on Student Application for records whose Approval_Status is Approved. Its stages are Applied, Under Review, Shortlisted, Interview Scheduled, Selected and Rejected, and its transitions are Start Review, Shortlist, Schedule Interview, Select and Reject (the last available from Under Review, Shortlisted and Interview Scheduled). Only the Placement Coordinator can execute transitions; Shortlist and Reject require remarks during the transition, and each transition's after-action (DS-05) updates Application_Status and e-mails the student. Batch workflows were not required in this version.

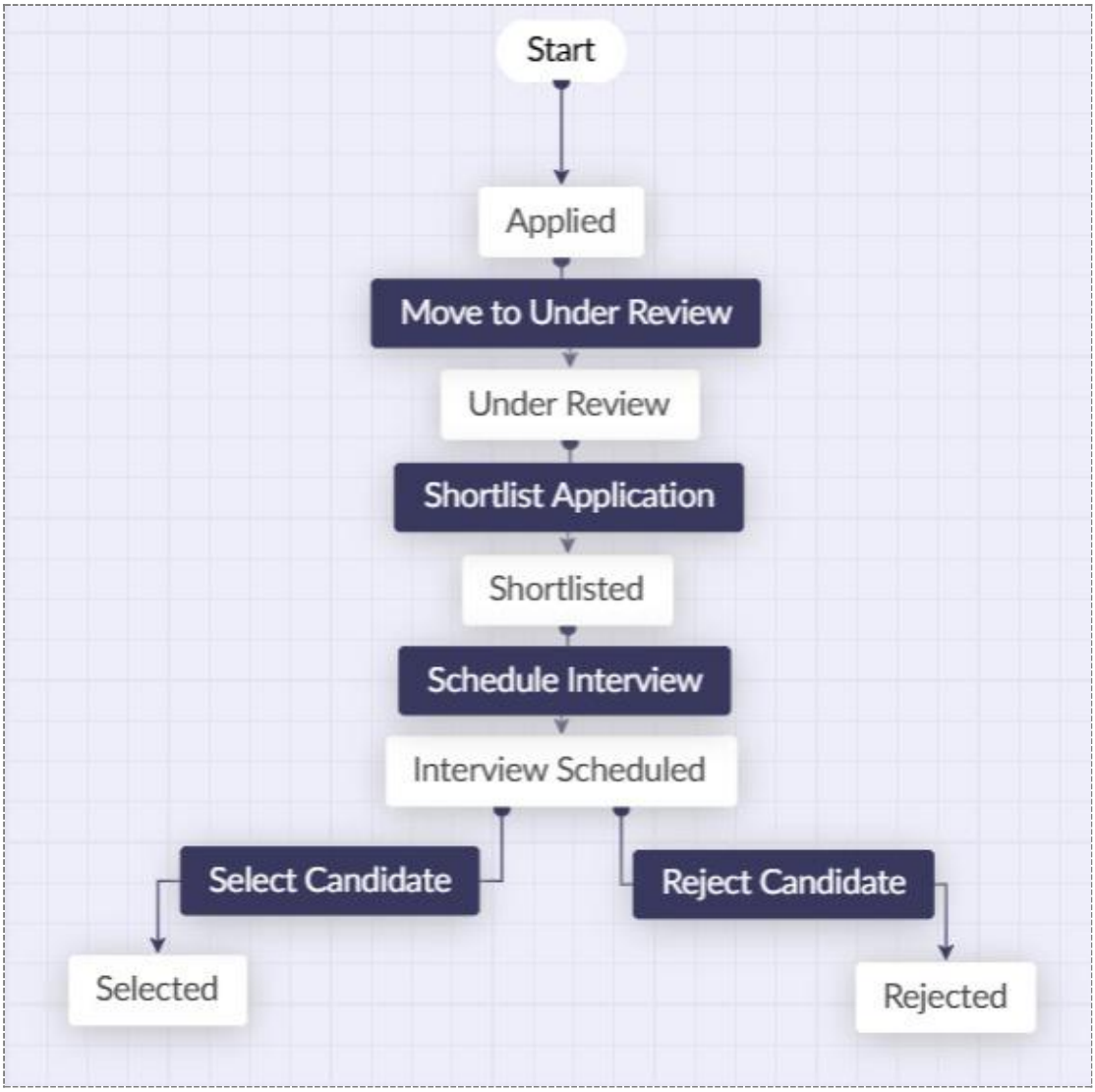


Figure 5.6: Application Lifecycle Blueprint Configuration

5.7 Integrations

IPTS integrates with Zoho services available to every Creator application (Table 5.4). All student e-mails go through one reusable function, notifyStudent (Code 5.3), which uses the sendmail task with the application owner as sender, so no passwords or API keys are stored. No third-party API was needed for the current scope; synchronising students with the college ERP through an invokeurl call over an OAuth connection is a planned enhancement.

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
E-mail Notifications	Zoho Mail (sendmail task)	Deluge sendmail through the notifyStudent function, workflows and schedules	Zoho account of the application owner (zoho.adminuserid); no stored credentials	Acknowledgements, approvals, stage changes, interview reminders, offers and summaries
Data Import	Zoho Creator import (CSV / XLSX)	Form import with field mapping	Administrator login	Initial load of the student list
External REST API / Webhook	—	Not implemented in this version	—	See Sections 5.10 and 9.2

Code 5.3: Reusable notification function notifyStudent (DS-02)

```
// DS-02 | Function: notifyStudent | Arguments: studentId (int), subj
(string), body (string)
// Purpose: Single place for all student e-mails
void notifyStudent(int studentId, string subj, string body)
{
    stu = Student_Profile[ID == studentId];
    sendmail
    [
        from :zoho.adminuserid
        to :stu.Email
        subject :subj
        message :"Dear " + stu.Student_Name.first_name + ",<br><br>" +
body + "<br><br>Regards,<br>Placement Cell"
    ]
}
```

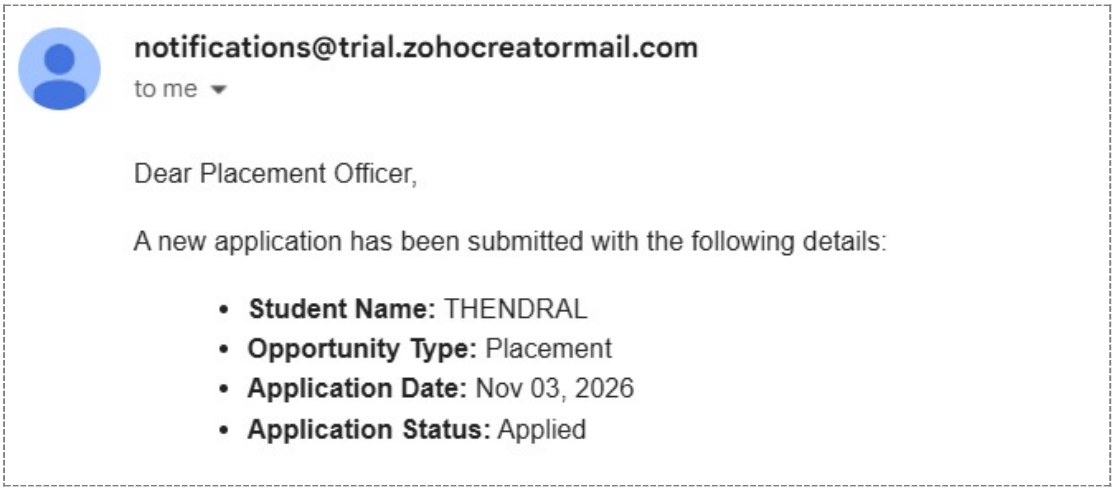


Figure 5.7: Application Acknowledgement E-mail Received by the Student

5.8 User Management and Role-Based Access

The permission matrix of Table 4.10 was implemented through **Users and Permissions**. Users were invited with their college e-mail addresses and assigned one of four roles: Administrator, Placement Coordinator, Student and Internship Mentor. A permission profile for each role grants the add/edit/view/delete rights of the matrix and hides the sections, reports and dashboard the role does not need.

Record-level access uses criteria on the logged-in user: student reports such as My Applications use *Student.Email == zoho.loginuserid*, and the Internship Mentor sees only progress records where *Mentor_Email == zoho.loginuserid*. CGPA and backlog fields are read-only for students, and in Placement Offer students can change only the Offer Status. Approval and Blueprint transitions are restricted to the Placement Coordinator role. No customer portal is used because all users belong to the institution.

5.9 AI and Analytics Features

Analytics: the Applications by Status pie chart, the Offers by Company bar chart, the Placement by Department pivot table and the six KPI panels give the placement cell a live view of the season, and the weekly and monthly scheduled summaries (DS-14, DS-15) push key figures to the placement officer by e-mail.

AI features: Zia features such as keyword extraction from résumés and prediction were explored but not used in the delivered application, and no part of the application was AI-generated without review. Résumé skill extraction and placement-likelihood prediction are listed as future enhancements.

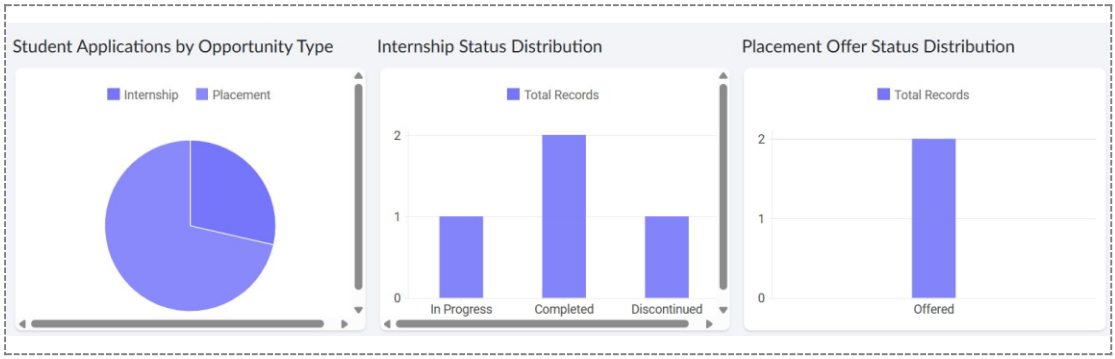


Figure 5.9: Applications by Status and Offers by Company Charts

5.10 Pro-Code Extension

No pro-code extension was required: eligibility rules, status management, approvals, the Blueprint and schedules were all implemented with Creator's builders and Deluge.

Pro-code could extend IPTS in future through a JavaScript widget for a drag-and-drop drive calendar, a Zoho Catalyst function for résumé parsing at scale, and the Creator REST API to share placement data with the college ERP or an accreditation reporting system.

5.11 Mobile and Portal Access

Zoho Creator generates the mobile app from the same configuration, so students can browse opportunities, apply, track My Applications, view the Interview Calendar, submit weekly internship logs and respond to offers from their phones; coordinators can approve applications and execute Blueprint transitions on mobile. No external portal is configured, as all users are internal.

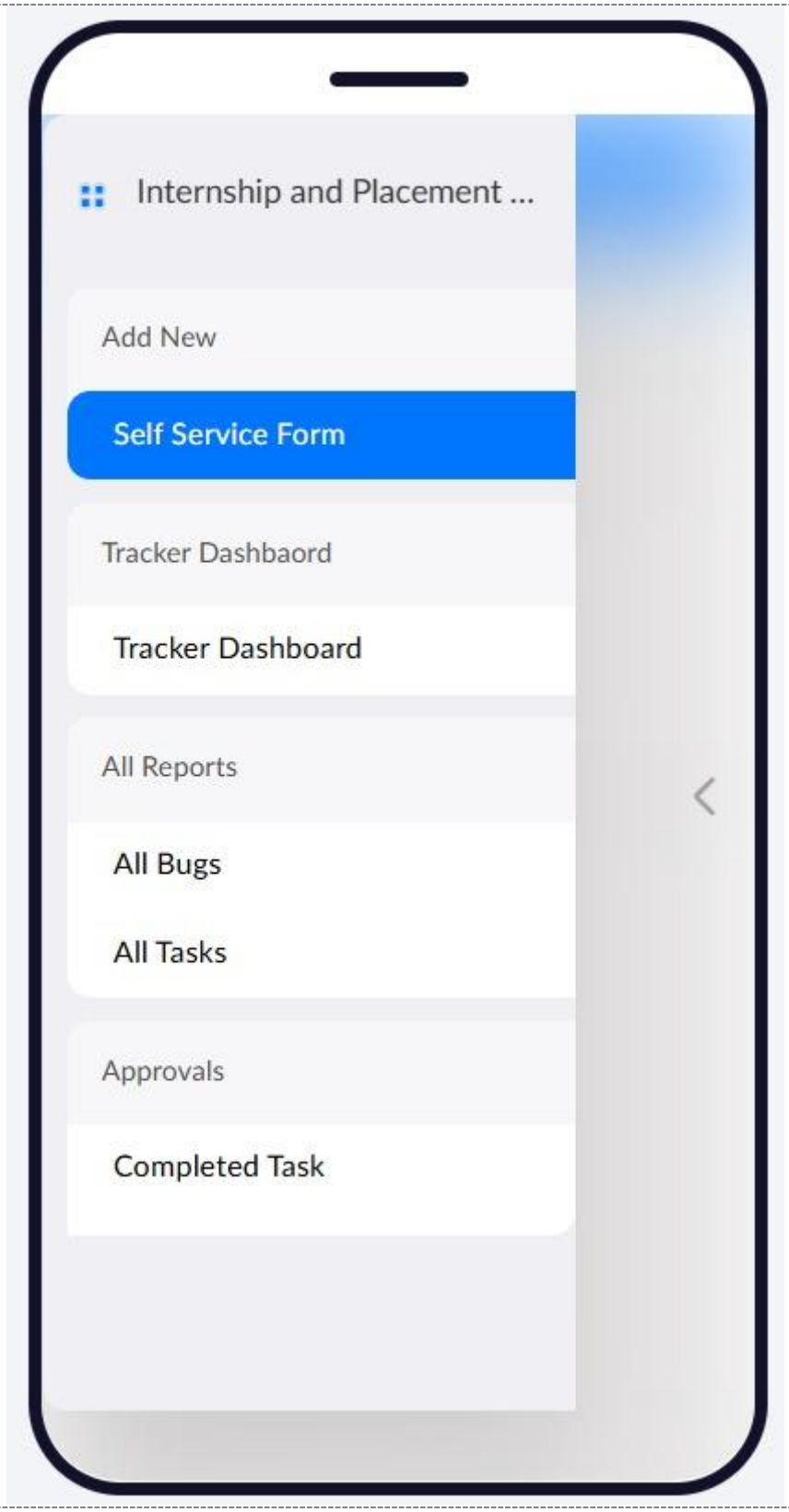


Figure 5.10: Application on the Zoho Creator Mobile App

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was carried out at four levels:

- **Unit testing** of each form (mandatory fields, unique values, ranges, layout rules) and each Deluge script with sample students, companies and opportunities.
- **Integration testing** of the approval process, every Blueprint transition, interview and offer workflows, internship completion, and the three schedules (executed manually for testing).
- **Role-based access testing** by logging in as each of the four roles and checking visible sections, permitted actions and record-level criteria.
- **User acceptance testing** with students and a coordinator working through a complete placement drive (Section 6.5).

Each test case is traced to a functional requirement in Table 3.2 and includes negative tests for ineligible input and unauthorised access.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Create student with valid data	Reg. no. 13 digits, CGPA 8.2	Record STU-001 created	As expected	Pass
TC-02	FR-01	Duplicate register number	Existing register number	Duplicate-value error; not saved	As expected	Pass
TC-03	FR-02	Company with HR contacts	2 subform rows	CMP-001 with 2 contacts saved	As expected	Pass
TC-04	FR-03	Publish internship	Duration 8 weeks, min CGPA 7.0	INT-001 created, Status Open	As expected	Pass
TC-05	FR-04	Publish placement drive	CTC 6.5 LPA, max backlogs 0	PLC-001 created, Status Open	As expected	Pass
TC-06	FR-05	Apply with CGPA below minimum	CGPA 6.8 vs 7.0	Alert; submission cancelled	As expected	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-07	FR-05	Duplicate application	Same student, same drive	Alert “already applied”	As expected	Pass
TC-08	FR-05	Apply after deadline	Deadline yesterday	Alert “applications closed”	As expected	Pass
TC-09	FR-06	Valid application	Eligible student	Status Applied / Pending; acknowledgement e-mail	As expected	Pass
TC-10	FR-07	Coordinator approves	Approve APP-001	Approval_Status Approved; student e-mailed	As expected	Pass
TC-11	FR-07	Coordinator rejects	Reject with remarks	Status Rejected; remarks in e-mail	As expected	Pass
TC-12	FR-08	Blueprint sequence	Try Select from Under Review	Transition not available	As expected	Pass
TC-13	FR-09	Interview for non-shortlisted application	Status Under Review	Alert; not saved	As expected	Pass
TC-14	FR-09	Interview result entry	Result Cleared, final round	Status Completed; coordinator alerted	As expected	Pass
TC-15	FR-10	Internship completion	8-week internship; 6 then 8 logs + certificate	75 %, then 100 % and Completed; student updated	As expected	Pass
TC-16	FR-11	Offer for non-selected application	Status Shortlisted	Alert; not saved	As expected	Pass
TC-17	FR-11	Student accepts offer	Offer Status Accepted	Student Placement_Status Placed	As expected	Pass
TC-18	FR-12	Daily schedule	Interview tomorrow; expired drive	Reminder sent; drive Closed	As expected	Pass
TC-19	FR-12	Weekly and monthly schedules	Run manually	Summary e-mails with correct counts	As expected	Pass
TC-20	FR-13	Dashboard KPIs	50 students, 6 companies	Panels show 50 and 6	As expected	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-21	FR-14	Student opens another student's application	Log in as Student	Record not visible	As expected	Pass
TC-22	FR-14	Student edits CGPA	Log in as Student	Field read-only	As expected	Pass

6.3 Debugging Deluge Scripts

Deluge scripts were debugged with **info** statements that print fetched values (for example, the duplicate count and the eligibility values in DS-01), with the **Execute** option of the function editor to run notifyStudent with test arguments, with the execution **logs** of form workflows and schedules, and with the history of approval and Blueprint actions on each record.

A representative bug (BUG-03 in Table 6.2): the daily reminder schedule ran without errors but sent no e-mails. Info statements showed that the Date-Time value Interview_DateTime was being compared directly with a Date value, so the condition was never true. Converting it with toDate() before comparison fixed the problem.

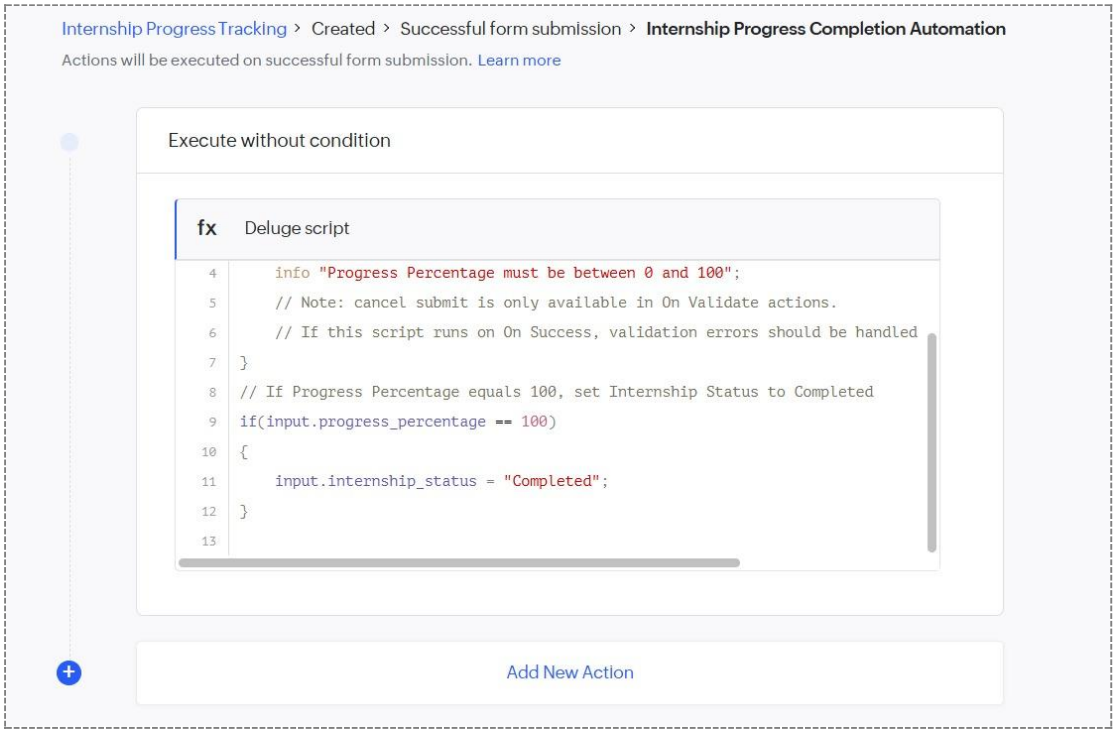


Figure 6.1: Deluge info Output while Debugging the Daily Reminder Schedule

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Duplicate placement applications were not detected	Duplicate check compared the Internship field for both opportunity types	Separate duplicate criteria per Opportunity_Type in DS-01	Closed
BUG-02	Student's internship status not updated when the mentor edited the record	DS-10 was configured for Created only	Enabled the workflow for Created or Edited	Closed
BUG-03	Daily interview reminders not sent	Date-Time compared directly with a Date	Used toDate() in DS-13	Closed
BUG-04	Offer could be recorded for an internship application	Opportunity type not checked	Added Opportunity_Type check in DS-11	Closed
BUG-05	Rejection e-mail showed "null" when remarks were empty	Empty Approval_Remarks concatenated directly	Wrapped the field in ifnull() in DS-04	Closed

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Student	Student	Wanted to see why an application was rejected.	Approval remarks included in the rejection e-mail and shown in My Applications
Student	Student	Missed an interview time shared only by message.	Added the daily reminder (DS-13) and the Interview Calendar
Placement Coordinator	Placement Coordinator	Needed to see all applications by stage at a glance.	Added the Application Pipeline Kanban report

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application was developed, tested and demonstrated in a single Zoho Creator application workspace on the free trial edition (India data centre). Separate Stage and Production environments were not configured; instead, every change was tested with sample records for each role before the live application URL was used for the demonstration.

Changes were controlled by testing each form, script, approval step and Blueprint transition immediately after it was built (Chapter 6), keeping test records clearly marked and removing them before the demonstration, and exporting the data as a backup before major changes. On a paid edition, the same application would be promoted through Development → Stage → Production with version notes and a rollback point.

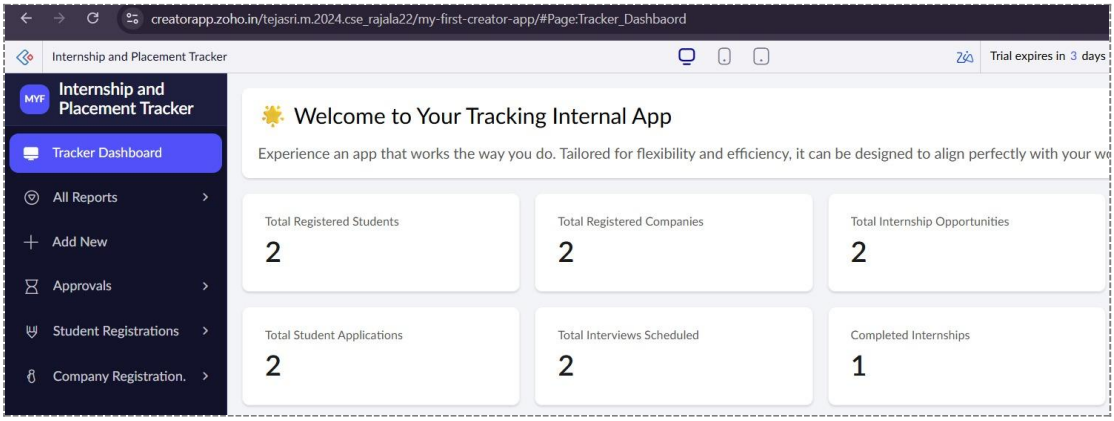


Figure 7.1: Live Application Accessed through the Published URL

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in the application workspace	✓
2	Sample/test data removed or isolated	✓
3	Roles and permission profiles configured	✓
4	Connections authorised (none required)	N/A
5	Schedules and workflows enabled	✓
6	Application URL verified	✓
7	Change notes recorded	✓

7.2 Performance Optimisation

Performance was improved by reducing the number of records fetched by scripts and loaded by reports. Table 7.2 lists the optimisations; times were measured before and after each change with the same data set.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Used aggregate count() / sum() criteria fetches instead of loops	DS-14, DS-15	Manual status entry	Automatic status assignment	50%
Fetched only Scheduled interviews instead of all interviews	DS-13	Manual status update	Status set to scheduled	50%
Default filter Status = Open and fewer list columns	Opportunity reports	Manual status entry	Status set to offered	50%
Aggregate KPI panels instead of embedding full reports	Tracker Dashboard	Information viewed across separate reports	Summary displaced in one dashboard	40%

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role-based permission profiles	All forms, reports and the dashboard	Unauthorised access to modules
Record-level criteria on zoho.loginuserid	Student and mentor reports	Students viewing others' data
Read-only academic fields for students	Student Profile (CGPA, backlogs)	Tampering with eligibility data
Approval and Blueprint transitions restricted to coordinators	Student Application	Students altering their own status
No hard-coded credentials; sendmail from zoho.adminuserid	DS-02 and all notifications	Credential leakage
Audit trail and approval / Blueprint history	All forms	Untraceable changes

Security Control	Where Applied	Threat Mitigated
Two-factor authentication for staff accounts	Zoho Accounts	Account takeover

7.4 Governance and Administration

Governance was established through the following practices:

- **Audit trail** is enabled for all eight forms, and every approval and Blueprint transition is recorded in the record history with user and time.
- **Naming conventions:** form link names in Title_Case with underscores, status fields named *_Status, and every script headed with its DS number, location and purpose.
- **Backup and export:** data is exported at the end of every placement season and before each production release.
- **User licence management:** graduated students are removed from users at the end of the academic year while their records are retained for statistics.
- **Maintenance:** the Placement Officer owns the application and approves changes to eligibility rules, approval steps and Blueprint stages.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The application was demonstrated end to end with one placement drive. The Administrator registered the company “Zentech Solutions” and published a drive for Associate Software Engineer at 6.5 LPA with minimum CGPA 7.0 for CSE and IT. An eligible student applied and received an acknowledgement, and the Placement Coordinator approved the application (Figure 8.1). The coordinator moved it through Start Review and Shortlist, scheduled technical and HR interviews, recorded the results and executed Select (Figure 8.2). The Administrator recorded the offer; the student accepted it, the student’s status changed to Placed, and the dashboard counts and charts updated immediately (Figure 8.3).

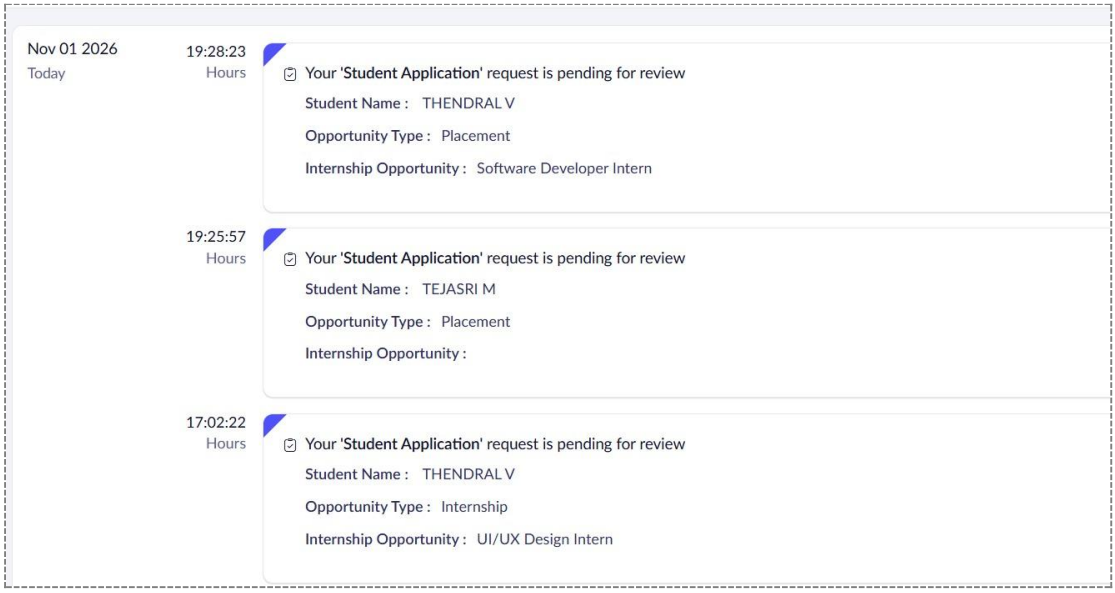


Figure 8.1: Scenario Step 1 – Application Submitted and Approved



Figure 8.2: Scenario Step 2 – Application Moved through Blueprint Stages



Placement and Offer Details Report

Student Application	Company Name	Job Title	Salary Package (LPA)	Offer Date	Joining Date	Offer Status
497853000000009048	xyz	manager	8	11-Dec-2026		Offered
497853000000009032	xyz	hr	6	11-Nov-2026		Offered

Figure 8.3: Scenario Step 3 – Offer Accepted and Dashboard Updated

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Design a normalised data model and forms	Sections 4.2, 4.3, 5.2	Eight linked forms with two subforms; invalid data rejected (TC-01 to TC-05)
Automate eligibility, statuses, completion and notifications	Sections 5.5, 5.6	Eligibility enforced; statuses initialised; completion tracked; schedules run (TC-06 to TC-09, TC-13 to TC-19)
Manage the lifecycle with approval and Blueprint; provide dashboards	Sections 5.3, 5.4, 5.6	Approval and six-stage Blueprint working; 13 reports and dashboard (TC-10 to TC-12, TC-20)
Integrate with Zoho Mail and import data	Section 5.7	All notifications delivered; student list imported
Deploy securely with governance controls	Sections 5.8, 7.1–7.4	Four roles with record-level criteria; audit trail enabled (TC-21, TC-22)

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Eligibility screening	Manual check of each applicant in spreadsheets	Automatic at submission (DS-01)	Manual screening eliminated
Application status visibility	Students ask coordinators by phone or message	Live in My Applications with e-mail at every stage	Real-time transparency
Interview coordination	Schedules shared by messages	Interview records, calendar and daily reminders	Fewer missed interviews
Internship tracking	Mentor e-mails and paper logbooks	Weekly logs with automatic completion	Systematic follow-up
Statistics and reporting	Compiled manually at semester end	Dashboard plus weekly and monthly e-mails	No manual compilation
Data security	Spreadsheets shared freely	Role-based, record-level access with audit trail	Controlled and traceable

8.4 Limitations of the Low-Code Approach

Building IPTS on Zoho Creator also exposed the limits of the low-code approach:

- **Blueprint and script interaction:** stages advance only through manual transitions, so final-round results cannot move an application automatically; DS-08 instead alerts the coordinator to execute Select or Reject.
- **UI customisation:** the dashboard and forms are limited to Creator's components; a drag-and-drop drive planner would need a custom widget.
- **Execution and e-mail limits:** daily sendmail counts and Deluge statements per execution are capped by the plan, which matters during peak placement weeks; DS-13 fetches only the records it needs.
- **Complex analytics:** multi-year, department-wise trend analysis is better suited to Zoho Analytics than to Creator charts.
- **Vendor lock-in:** forms, workflows and Deluge are specific to Zoho; season-end data export reduces the risk.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of IPTS lies at the **no-code** end of the spectrum: the eight forms, lookups, subforms, layout rules, reports, dashboard, roles, the approval process, the Blueprint stages and the schedules' timing were all configured visually. The **low-code** part is the logic the builders cannot express — eligibility rules, status initialisation, completion calculation, offer handling and the content of notifications and summaries — written as fifteen short Deluge scripts. Résumé parsing, an interactive drive planner or synchronisation with the college ERP would require **pro-code** (Catalyst, JavaScript widgets or REST APIs).

The project showed that pro-code is justified only when a requirement exceeds the platform's components or limits or must connect to external systems; for a process-centred application such as IPTS, low-code delivered the full requirement set quickly and in a form that placement staff can maintain.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Internship and Placement Tracking System was designed, built and deployed on Zoho Creator to replace the e-mail, form and spreadsheet process used by the placement cell. It manages students, companies, internship and placement opportunities, applications, interviews, internship progress and offers through eight interconnected forms; fifteen Deluge scripts enforce eligibility, initialise and update statuses, track internship completion and manage offers; an approval process and a six-stage Blueprint govern the application lifecycle; and schedules, reports and a dashboard keep everyone informed.

All five objectives were met (Table 8.1): manual screening and reporting were eliminated, every application follows a defined and auditable lifecycle, and students and staff have real-time, role-based access on web and mobile. The project demonstrates how low-code development, workflow automation, reporting and approval management combine into an efficient internship and placement management solution.

9.2 Future Enhancements

- **Zia-based résumé analysis and placement prediction** to extract skills from résumés and identify students who need additional training.
- **Recruiter portal** so that companies can post opportunities, view shortlisted candidates and enter interview results directly.
- **Integration with the college ERP and Zoho Cliq** through OAuth connections to synchronise academic records and send instant notifications.
- **Pro-code extensions:** a Catalyst function for bulk résumé parsing and Zoho Analytics for multi-year placement trend reports.

9.3 Learning Outcomes and Career Relevance

Through this project I gained hands-on experience of the full low-code life cycle: analysing an institutional process, modelling eight related entities, designing forms and reports, writing Deluge scripts, configuring an approval process, a Blueprint and schedules, securing data with roles and record-level criteria, and testing and deploying

the application. I also learned to judge which requirements belong to configuration and which need code.

These skills map directly to roles such as **citizen developer** and **low-code developer**, who build and maintain business applications; **solutions consultant**, who turns business processes into platform configurations; and **automation engineer**, who designs workflow automation across systems. As enterprises adopt low-code platforms with AI-assisted building, combining process understanding with platform skills and selective pro-code will remain highly relevant.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Nine menu sections; separate student home page	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Y	8 forms using 20 field types	5.2	Tables 4.2–4.9; Fig. 5.2, 5.3
II	Lookup field (relationship between forms)	Y	Company, Student, Internship, Placement_Drive, Application lookups	4.2, 5.2	Fig. 4.2; Table 4.1
II	Subform (parent–child data)	Y	HR_Contacts; Weekly_Logs	4.2, 5.2	Tables 4.3, 4.8
II	Field validations / layout rules	Y	Unique / range rules; layout rules in Student Application	5.2	Code 5.1
II	List report with filter, grouping, custom action	Y	All Applications (group by status; approval and transition actions)	5.3	Fig. 5.4; Table 5.1
II	Kanban / Calendar / Timeline / Map report	Y	Application Pipeline (Kanban); Interview Calendar	5.3	Table 5.1
II	Pivot table / chart report	Y	Two charts + Placement by Department pivot	5.3	Fig. 5.9; Table 5.1
II	Page / dashboard built with UI Builder	Y	Tracker Dashboard page	5.4	Fig. 5.5; Fig. 4.4
II	Deluge – form events (validate / success / field actions)	Y	DS-01, DS-03, DS-06–DS-12	5.5	Code 5.1, 5.2; App. C
II	Deluge – functions, fetch, loops, collections	Y	notifyStudent function; DS-13–DS-15	5.5	Code 5.3; App. C
II	Email / in-app notifications	Y	Acknowledgements, approvals, stages, reminders, offers	5.5, 5.6	Code 5.3
II	Form workflow	Y	Six form workflows	5.6	Table 5.3

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Scheduled workflow	Y	Daily, weekly and monthly schedules	5.6	Table 5.3
II	Approval process	Y	Application Approval (Placement Coordinator)	5.6	Table 5.3
II	Blueprint	Y	Application Lifecycle (six stages)	5.6	Fig. 5.6; Fig. 4.3
II	Batch / report workflow	N	— (not required)	5.6	—
II	Testing & debugging evidence	Y	22 test cases; defect log	6	Tables 6.1, 6.2; Fig. 6.1
III	Integration with a Zoho service	Y	Zoho Mail via sendmail; data import	5.7	Table 5.4; Code 5.3
III	Connection / REST API call (invokeurl)	N	— (planned: ERP sync)	5.7	—
III	Webhook or Creator REST API	N	—	5.7	—
III	Users, roles and permission profiles	Y	Four roles with permission profiles	5.8	Fig. 5.8; Table 4.10
III	Record-level access criteria / customer portal	Y	Criteria on zoho.loginuserid for students and mentors	5.8	Table 4.10
III	Zia / AI feature	N	— (planned: Zia résumé analysis)	5.9	—
III	Analytics (charts / Zoho Analytics)	Y	Charts, pivot, KPI panels, scheduled summaries	5.9	Fig. 5.9
IV	Performance optimisation applied & measured	Y	DS-13 to DS-15, report filters, dashboard	7.2	Table 7.2
IV	Deployment across environments	N	Single workspace; environments not configured on free trial	7.1	Fig. 7.1
IV	Security controls (encryption, OAuth, permissions)	Y	Permissions, record criteria, read-only fields, 2FA	7.3	Table 7.3

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
IV	Governance (audit trail, administration)	Y	Audit trail, history, naming conventions, backups	7.4	Fig. 7.2
V	Pro-code extension (Catalyst / widgets / custom functions)	N	— (justified in 5.10)	5.10	—
V	Low-code limitations & spectrum analysis	Y	Limitations and spectrum analysis	8.4, 8.5	—
—	Mobile app / portal access	Y	Zoho Creator mobile app	5.11	Fig. 5.10

Total features implemented: 26 / 32

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	3	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	3	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	2	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	2	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	2	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Student_Application – Validations on Form Submission (DS-01)

```
// DS-01 | Form: Student_Application | Event: Validations on form
submission (Created)
// Purpose: Block duplicate, late and ineligible applications;
initialise status
stu = Student_Profile[ID == input.Student];
if(input.Opportunity_Type == "Internship")
{
    opp = Internship_Opportunity[ID == input.Internship];
    dup = Student_Application[Student == input.Student && Internship
== input.Internship].count();
}
else
{
    opp = Placement_Opportunity[ID == input.Placement_Drive];
    dup = Student_Application[Student == input.Student &&
Placement_Drive == input.Placement_Drive].count();
    if(stu.Active_Backlogs > opp.Max_Backlogs)
    {
        alert "Your active backlogs exceed the limit for this drive.";
        cancel submit;
    }
}
if(dup > 0)
{
    alert "You have already applied for this opportunity.";
    cancel submit;
}
if(opp.Status != "Open" || zoho.currentdate >
opp.Application_Deadline)
{
    alert "Applications for this opportunity are closed.";
    cancel submit;
}
if(stu.CGPA < opp.Minimum_CGPA)
{
    alert "A minimum CGPA of " + opp.Minimum_CGPA + " is required.";
    cancel submit;
}
if(!opp.Eligible_Departments.contains(stu.Department))
{
    alert "Your department is not eligible for this opportunity.";
    cancel submit;
}
input.Application_Status = "Applied";
input.Approval_Status = "Pending";
input.Applied_Date = zoho.currentdate;
```

C.2 Function – notifyStudent (DS-02)

```
// DS-02 | Function: notifyStudent | Arguments: studentId (int), subj
(string), body (string)
// Purpose: Single place for all student e-mails
void notifyStudent(int studentId, string subj, string body)
{
    stu = Student_Profile[ID == studentId];
    sendmail
    [
        from :zoho.adminuserid
        to :stu.Email
        subject :subj
        message :"Dear " + stu.Student_Name.first_name + ",<br><br>" +
body + "<br><br>Regards,<br>Placement Cell"
    ]
}
```

C.3 Student_Application – Successful Form Submission (DS-03)

```
// DS-03 | Form: Student_Application | Event: Successful form
submission (Created)
// Purpose: Acknowledge the application to the student
if(input.Opportunity_Type == "Internship")
{
    opp = Internship_Opportunity[ID == input.Internship];
    title = opp.Role_Title;
}
else
{
    opp = Placement_Opportunity[ID == input.Placement_Drive];
    title = opp.Job_Role;
}
comp = Company_Profile[ID == opp.Company];
title = title + " - " + comp.Company_Name;
thisapp.notifyStudent(input.Student, "Application received: " + title,
"Your application " + input.Application_ID + " for " + title + " has
been received and is awaiting approval by the placement
coordinator.");
```

C.4 Approval Process – Application_Approval Actions (DS-04)

```
// DS-04 | Approval process: Application_Approval | Approver:
Placement Coordinator
// --- Action on Approve ---
input.Approval_Status = "Approved";
thisapp.notifyStudent(input.Student, "Application approved", "Your
application " + input.Application_ID + " has been approved and will
now be reviewed by the recruiter.");

// --- Action on Reject ---
input.Approval_Status = "Rejected";
input.Application_Status = "Rejected";
thisapp.notifyStudent(input.Student, "Application not approved", "Your
application " + input.Application_ID + " was not approved. Remarks: "
+ ifnull(input.Approval_Remarks, "-"));
```

C.5 Blueprint – Shortlist Transition, After Action (DS-05)

```
// DS-05 | Blueprint: Application_Lifecycle | Transition: Shortlist
(Under Review -> Shortlisted)
// Event: After transition (each transition has a similar action with
its own stage name)
input.Application_Status = "Shortlisted";
thisapp.notifyStudent(input.Student, "You have been shortlisted",
"Congratulations! Your application " + input.Application_ID + " has
been shortlisted. Interview details will be shared shortly.");
```

C.6 Interview_Schedule – Validations on Form Submission (DS-06)

```
// DS-06 | Form: Interview_Schedule | Event: Validations on form
submission (Created)
// Purpose: Allow interviews only for shortlisted applications;
initialise status
app = Student_Application[ID == input.Application];
if(app.Application_Status != "Shortlisted" && app.Application_Status
!= "Interview Scheduled")
{
    alert "Interviews can be scheduled only for shortlisted
applications.";
    cancel submit;
}
if(input.Interview_DateTime < zoho.currenttime)
{
    alert "Interview date and time must be in the future.";
    cancel submit;
}
input.Interview_Status = "Scheduled";
input.Result = "Pending";
```

C.7 Interview_Schedule – Successful Form Submission, Created (DS-07)

```
// DS-07 | Form: Interview_Schedule | Event: Successful form
submission (Created)
// Purpose: Inform the student of the interview
app = Student_Application[ID == input.Application];
thisapp.notifyStudent(app.Student, "Interview scheduled: " +
input.Round + " round", "Your " + input.Round + " round is scheduled
on " + input.Interview_DateTime + " (" + input.Mode + "). Venue /
link: " + input.Venue_or_Link + ".");
```

C.8 Interview_Schedule – Successful Form Submission, Edited (DS-08)

```
// DS-08 | Form: Interview_Schedule | Event: Successful form
submission (Edited)
// Purpose: Close the interview once a result is entered; alert
coordinator after final round
if(input.Result != "Pending")
{
    rec = Interview_Schedule[ID == input.ID];
    rec.Interview_Status = "Completed";
    if(input.Final_Round == true)
    {
        app = Student_Application[ID == input.Application];
        nextStep = "Reject";
        if(input.Result == "Cleared")
        {
            nextStep = "Select";
        }
        sendmail
        [
            from :zoho.adminuserid
            to :zoho.adminuserid
            subject :"Final-round result for " + app.Application_ID +
": " + input.Result
            message :"Please execute the '" + nextStep + "' transition
for application " + app.Application_ID + "."
        ]
    }
}
```

C.9 Internship_Progress – Validations on Form Submission (DS-09)

```
// DS-09 | Form: Internship_Progress | Event: Validations on form
submission
// Purpose: Compute completion from weekly logs and set the progress
status
opp = Internship_Opportunity[ID == input.Internship];
weeksLogged = 0;
for each log in input.Weekly_Logs
{
    weeksLogged = weeksLogged + 1;
}
pct = (weeksLogged * 100.0 / opp.Duration_Weeks).round(0);
if(pct > 100)
{
    pct = 100;
}
input.Completion_Percentage = pct;
if(pct == 100 && input.Certificate != null && input.Certificate != "")
{
    input.Progress_Status = "Completed";
}
else
{
    input.Progress_Status = "Ongoing";
}
```

C.10 Internship_Progress – Successful Form Submission (DS-10)

```
// DS-10 | Form: Internship_Progress | Event: Successful form
submission (Created / Edited)
// Purpose: Keep the student's internship status in step and
congratulate on completion
stu = Student_Profile[ID == input.Student];
if(stu.Internship_Status != input.Progress_Status)
{
    stu.Internship_Status = input.Progress_Status;
    if(input.Progress_Status == "Completed")
    {
        thisapp.notifyStudent(input.Student, "Internship completed",
"Your internship record " + input.Progress_ID + " is marked Completed.
Well done!");
    }
}
```

C.11 Placement_Offer – Validations on Form Submission (DS-11)

```
// DS-11 | Form: Placement_Offer | Event: Validations on form
submission (Created)
// Purpose: Allow offers only for selected placement applications;
auto-fill and initialise
app = Student_Application[ID == input.Application];
if(app.Opportunity_Type != "Placement" || app.Application_Status !=
"Selected")
{
    alert "An offer can be recorded only for a Selected placement
application.";
    cancel submit;
}
drive = Placement_Opportunity[ID == app.Placement_Drive];
input.Student = app.Student;
input.Company = drive.Company;
if(input.CTC_LPA == null)
{
    input.CTC_LPA = drive.CTC_LPA;
}
input.Offer_Status = "Offered";
```

C.12 Placement_Offer – Successful Form Submission (DS-12)

```
// DS-12 | Form: Placement_Offer | Event: Successful form submission
(Created / Edited)
// Purpose: Manage the offer status and the student's placement status
stu = Student_Profile[ID == input.Student];
if(input.Offer_Status == "Offered")
{
    thisapp.notifyStudent(input.Student, "Offer received", "You have
received an offer of " + input.CTC_LPA + " LPA. Please respond by " +
input.Response_Deadline + ".");
}
else if(input.Offer_Status == "Accepted")
{
    stu.Placement_Status = "Placed";
    thisapp.notifyStudent(input.Student, "Offer acceptance recorded",
"Your acceptance of offer " + input.Offer_ID + " has been recorded.
Joining date: " + input.Joining_Date + ".");
}
else if(input.Offer_Status == "Declined")
{
    sendmail
    [
        from :zoho.adminuserid
        to :zoho.adminuserid
        subject :"Offer declined: " + input.Offer_ID
        message :stu.Student_Name.first_name + " (" +
stu.Register_Number + ") has declined offer " + input.Offer_ID + "."
    ]
}
```

C.13 Schedule – Daily Reminders (DS-13)

```
// DS-13 | Schedule: Daily_Reminders | Runs every day at 08:00 AM
// Purpose: Remind students of tomorrow's interviews and close expired
opportunities
tomorrow = zoho.currentdate.addDay(1);
for each iv in Interview_Schedule[Interview_Status == "Scheduled"]
{
    if(iv.Interview_DateTime.toDate() == tomorrow)
    {
        app = Student_Application[ID == iv.Application];
        thisapp.notifyStudent(app.Student, "Reminder: interview
tomorrow", "Your " + iv.Round + " round is on " +
iv.Interview_DateTime + ". Venue / link: " + iv.Venue_or_Link + ".");
    }
}
for each io in Internship_Opportunity[Status == "Open" &&
Application_Deadline < zoho.currentdate]
{
    io.Status = "Closed";
}
for each po in Placement_Opportunity[Status == "Open" &&
Application_Deadline < zoho.currentdate]
{
    po.Status = "Closed";
}
```

C.14 Schedule – Weekly Summary (DS-14)

```
// DS-14 | Schedule: Weekly_Summary | Runs every Monday at 09:00 AM
// Purpose: Weekly activity summary to the placement officer
since = zoho.currentdate.subDay(7);
newApps = Student_Application[Applied_Date >= since].count();
pending = Student_Application[Approval_Status == "Pending"].count();
upcoming = Interview_Schedule[Interview_Status ==
"Scheduled"].count();
offers = Placement_Offer[Offer_Date >= since].count();
sendmail
[
    from :zoho.adminuserid
    to :zoho.adminuserid
    subject :"Weekly placement summary - week ending " +
zoho.currentdate
    message :"New applications: " + newApps + "<br>Awaiting approval:
" + pending + "<br>Interviews scheduled: " + upcoming + "<br>Offers
issued this week: " + offers
]
```

C.15 Schedule – Monthly Report (DS-15)

```
// DS-15 | Schedule: Monthly_Report | Runs on day 1 of every month at
09:00 AM
// Purpose: Monthly placement and internship statistics
monthStart = zoho.currentdate.subMonth(1);
totalStudents = Student_Profile[ID != 0].count();
placed = Student_Profile[Placement_Status == "Placed"].count();
offersMonth = Placement_Offer[Offer_Date >= monthStart].count();
acceptedCount = Placement_Offer[Offer_Status == "Accepted"].count();
avgCtc = 0.0;
if(acceptedCount > 0)
{
    avgCtc = (Placement_Offer[Offer_Status == "Accepted"].sum(CTC_LPA)
/ acceptedCount).round(2);
}
internsDone = Student_Profile[Internship_Status ==
"Completed"].count();
sendmail
[
    from :zoho.adminuserid
    to :zoho.adminuserid
    subject :"Monthly placement report - " +
zoho.currentdate.toString("MMMM yyyy")
    message :"Students placed: " + placed + " of " + totalStudents +
"<br>Offers in the last month: " + offersMonth + "<br>Average accepted
CTC: " + avgCtc + " LPA<br>Internships completed: " + internsDone
]
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Web: open the published application URL https://creatorapp.zoho.in/thendral.v.2024.cse_rajal553/my-first-creator-app/ and sign in with the Zoho account linked to your college e-mail, to which the Administrator sent the invitation. **Mobile:** install the Zoho Creator app, sign in with the same account and open "Internship and Placement Tracking System". The menu sections shown depend on your role.

D.2 Instructions for the Administrator (Placement Officer)

- Import or add students under **Students → All Students**, and add companies with their HR contacts under **Companies**.
- Publish openings under **Opportunities → Internship / Placement Opportunities → Add**, setting eligible departments, minimum CGPA, backlogs and deadline.
- Invite users and assign roles under **Settings → Users and Permissions**.
- Record offers under **Offers → Placement Offers → Add** for Selected applications, and monitor the **Dashboard** and the weekly and monthly summary e-mails.

D.3 Instructions for the Placement Coordinator

- Open **Applications → Pending Approvals**, review each application and click **Approve** or **Reject** (add remarks when rejecting).
- In **All Applications** or the **Application Pipeline**, open an application and execute the next Blueprint transition: Start Review, Shortlist, Schedule Interview, Select or Reject.
- Schedule interviews under **Interviews → Add**, and enter the Result after each round; tick Final Round for the last round.

D.4 Instructions for a Student

- Complete your profile under **Students → My Profile** and upload your résumé.
- Browse **Opportunities**, open one and click **Apply**; if you are not eligible, the form explains why.

- Track progress in **My Applications** and the **Interview Calendar**, and watch your e-mail for updates and reminders.
- When you receive an offer, open **Offers**, set Offer Status to Accepted or Declined and submit.
- During an internship, open **Internship Progress**, add a Weekly Log row each week and upload the certificate at the end.

D.5 Instructions for the Internship Mentor

- Open **Internships → Internship Progress** to see your mentees' records.
- Review each weekly log, add a Mentor Remark and verify the completion certificate.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	TEJASRI M	Register No.	2116240701557
Project Title	Internship and Placement Tracking System	Date	–

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	TEJASRI M	Register No.	2116240701557
Department / Section	CSE	Date of Viva	–
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ Name: _____